

Family In The Loop: Enabling Family Involvement and Person-Centered Dementia Care at Long-Term Care Facilities with Collaborative AI Tools

DYLAN EDWARD MOORE, Dartmouth College, USA

SONGYUN TAO, Dartmouth College, USA

EMMA RICCI-DE LUCCA, Dartmouth College, USA

CHRISTINA SAPP TADIN, Memcara Inc., USA

DIO TADIN, Memcara Inc., USA

BRIAN MORGAN, Memcara Inc., USA

ELIZABETH L MURNANE, Dartmouth College, USA

Person-centered care provides a critical framework for dementia support in long-term care facilities, emphasizing attention to each resident's evolving needs, personal history, preferences, and values. Achieving this level of personalized attention depends on coordination among staff and active family involvement to record and share this essential context. However, fragmented communication, caregiver burnout, limited resources, and regulatory complexity frequently disrupt the flow of information, creating conditions in which person-centered care becomes difficult to sustain in practice.

To address these challenges, we introduce Family In The Loop (FITL), an AI-enabled web platform that enhances continuity and collaboration for person-centered dementia care. FITL automatically synthesizes diverse care data—including video, notes, and records—into contextually relevant, emotionally resonant updates tailored for staff and family members. It offers four interlinked views: "Reels" provides families with concise, meaningful video updates; "Essentials" supplies caregivers with resident-specific summaries for immediate use; "Explanations" presents narrative insights into care decisions; and "Archives" maintains a searchable, longitudinal record of care activities. Together, these views enable flexible yet coordinated information sharing, respecting the fluid nature of family involvement and supporting continuity across caregiver roles and shifts.

Through an iterative co-creation process spanning two years and seven resource-constrained facilities, we designed, developed, and deployed FITL in real-world dementia care contexts. Findings indicate that FITL improved staff access to critical resident information and facilitated sustained family engagement without increasing caregiver burdens. By surfacing timely, personalized, and meaningful information, FITL strengthened trust between families and staff, reduced misunderstandings, and bridged communication gaps. Our work highlights the potential of AI tools to stitch fragmented care interactions into cohesive, person-centered caregiving ecosystems. We conclude the paper by discussing inherent tensions around automation, oversight, and privacy, and we propose design principles for adaptive, collaborative, and person-centered care technologies.

CCS Concepts: • **Human-centered computing** → **Human computer interaction (HCI); Collaborative and social computing systems and tools.**

Additional Key Words and Phrases: Dementia, Caregiving, Long-Term Care, Human-Centered Design, Person-Centered Care, AI, Human-Centered AI, Human-AI Interaction, Collaborative Health Technologies, Family-Centered Care

D Moore, 15 Thayer Dr Room 026, Hanover, NH 03755, USA; Memcara Inc., 58 Gretna Woods Road, Pleasant Valley, NY 12569; E Murnane, 15 Thayer Dr Room 026, Hanover, NH 03755, USA.



This work is licensed under a [Creative Commons Attribution-NonCommercial 4.0 International License](https://creativecommons.org/licenses/by-nc/4.0/).

© 2025 Copyright held by the owner/author(s).

ACM 2573-0142/2025/11-ARTCSCW418

<https://doi.org/10.1145/3757599>

ACM Reference Format:

Dylan Edward Moore, Songyun Tao, Emma Ricci-De Lucca, Christina Sapp Tadin, Dio Tadin, Brian Morgan, and Elizabeth L Murnane. 2025. Family In The Loop: Enabling Family Involvement and Person-Centered Dementia Care at Long-Term Care Facilities with Collaborative AI Tools. *Proc. ACM Hum.-Comput. Interact.* 9, Article CSCW418 (November 2025), 45 pages. <https://doi.org/10.1145/3757599>

1 Introduction

As the global population ages, the number of people living with dementia (PLWD) continues to rise, with estimates projecting an increase from 57 million in 2021 to 78 million by 2030 [42, 44]. Long-term care facilities, already stretched thin, are under increasing pressure to meet the complex and evolving needs of this population [50]. Unlike general geriatric care, dementia care involves navigating a wide range of neuropsychiatric symptoms—from cognitive decline and behavioral disturbances to progressive loss of function—that often lead to near-total dependence in the later stages of the disease [32, 33, 51].

Person-centered care is widely recognized as the gold standard in dementia care. It views the resident as a person with an enduring sense of self, unique preferences and personality, and seeks to support this personhood [36]. Evidence-based therapies such as reminiscence, validation, and cognitive stimulation reflect this approach and offer pathways to more meaningful engagement. However, in practice, these therapies are difficult to sustain. Delivering them requires time, emotional energy, and deep contextual knowledge—resources that are often scarce on the ground. Understaffing, burnout, and high turnover leave caregivers overextended and reactive, rather than proactive [23].

These constraints are further compounded by information breakdowns. As dementia progresses, PLWD lose the ability to explain what helps, what harms, or how they're feeling—placing even more pressure on staff to interpret subtle signs without consistent guidance. Meanwhile, gaps in documentation, siloed communication, and underutilized family knowledge disrupt continuity of care [10, 56]. The result is a persistent gap between what high-quality, person-centered care promises—and what the realities of care work allow.

Emerging digital interventions, such as mobile apps, web portals, and AI-driven assistants, offer promising avenues for meeting this need [14, 57]. These technologies can provide scalable solutions for real-time information sharing on resident status, care plans, and facility decisions, potentially transforming the ways families stay connected and informed [60]. For example, a well-designed digital portal could give families timely, accurate insights into their loved one's care, fostering trust, alleviating anxiety, and allowing them to feel actively involved in the decision-making process without imposing additional burdens on caregiving staff. In particular, digital agents, software tools designed to perform tasks, gather information, and interact with users either fully or semi-autonomously, have emerged as an active area of research in health care contexts, including long-term care facilities [54]. Such tools offer versatile solutions that help facilities manage complex workflows, support caregivers in resource-constrained facilities, and improve resident well-being [35, 53]. However, implementing such technologies for PLWD requires careful attention to unique challenges of this context and the condition itself, including the challenges of shared caregiving responsibilities, as well as privacy, legal, and ethical considerations [39].

Although existing dementia care models and digital interventions integrate biological, psychological, and social factors and acknowledge the crucial role of family members in the care of PLWD, they often overlook the role of the care recipient's family system and relationships with caregiving staff as determinants of care outcomes [45]. Moreover, little research has examined how technology-driven information-sharing affects family-facility dynamics. To address this gap, we adopt a more inclusive dementia care framework that emphasizes not only person-centered

care but also a family-centered lens [27, 45], which considers the relationships within the social ecological networks that encompass care recipients, families, and caregivers. Our paper specifically explores a gap in the literature on the design of artificial intelligence (AI) tools that promote family engagement in person-centered dementia caregiving at long term care facilities.

In this paper, we present the needfinding, design, development, and evaluation of **“Family In The Loop” (FITL)**, a web-based system that supports person-centered dementia care by bridging communication gaps between families and care staff. FITL supports family-staff collaboration by tailoring shared access to care information through four interlinked views—providing families with emotionally attuned visibility into the resident experience, while enabling caregivers to retrieve essential information on demand, all without disrupting existing workflows. By supporting the capture, curation, and sharing of care-related information, FITL addresses the entangled emotional, clinical, and contextual needs of users. Through a multi-year, multi-site collaboration, we identify staff and family-specific needs, navigate institutional concerns around documentation and liability, and evaluate a deployed prototype. This work contributes new insights into how sociotechnical systems can enhance visibility, trust, and collaboration in long-term dementia care.

Specifically, our efforts were guided by the following research questions that reflect that process:

- RQ1: What are the experiences of families and professional caregivers of care recipients in long-term care facilities, particularly with respect to challenges in communication and family involvement?
- RQ2: How might we design and build an interactive tool that addresses the needs of family members and caregivers and enhances relationship-building to foster a greater sense of engagement for families of PLWD?
- RQ3: How does the tool impact families’ sense of involvement in care as well as other caregivers’ practices?

2 Related Work

2.1 Person-Centered Dementia Care in Long-Term Care Facilities

Numerous medications exist to address the behavioral and psychological symptoms of dementia; however, these treatments often lead to unwanted side effects and have limited effectiveness [18, 31]. Overmedication of psychotropic drugs in particular is common among PLWD at care facilities and is a point of concern for families and staff, due to risks for adverse drug effects, drug interactions, and negative impacts on quality of life [21]. In response, there is a growing shift towards non-pharmacological interventions in dementia care, which prioritize cognitive training, physical activity, and behavioral techniques [7, 29]. These strategies aim to promote the abilities and potential of PLWD, avoiding a pathologized view of their condition. Consequently, we prefer the term “care recipients” over “patients” to reduce associations with pathology and deficit-based labeling.

Providing holistic care to PLWD in long-term care facilities is essential to improving their quality of life, including enhancing emotional well-being, social engagement, cognitive stimulation, and physical functioning [58]. *Person-centered care* goes beyond treating symptoms and emphasizes understanding a PLWD’s history, preferences and values to ensure compassionate and responsive practice [36]. Person-centered care is crucial in the context of dementia because it upholds the dignity of individuals and addresses their distinct cognitive and emotional requirements [20, 43]. This model improves interactions through person-centered communication [59], minimizes the dependence on potentially harmful antipsychotic medications for managing agitation [22], and significantly contributes to the overall well-being of both care recipients and caregivers [8]. By

strengthening interpersonal connections and alleviating stress, this approach facilitates a more supportive caregiving environment.

To deliver such care, prior research highlights the importance of respecting a care recipient's preferences, providing education and emotional support, and involving family members in their care [48]. Although caregivers have the desire to foster greater communication with care recipient's families, cumbersome documentation, staff shortages, high workload, and increased attrition rates render this challenging [28]. AI-driven digital health technology has the potential to provide valuable contributions towards mitigating the aforementioned problems by enhancing personalized communication and increasing support for caregivers [57].

In designing such tools, we note that care ecologies can be conceptualized as complex adaptive systems, offering a framework for comprehending the intricacies of care delivery, the inherent unpredictability, and the feedback mechanisms that influence the well-being of both caregivers and care recipients. This framing emphasizes the interactions and emergent behaviors among various agents, including caregivers, care recipients, families, administrators, and digital agents that are characterized by dynamic, non-linear behaviors, where agents adapt based on their interactions. These qualities mean that interventions, including digital agents, must similarly be adaptive to shifting contexts to accommodate the evolving nature of the environment and its users [11, 12, 25, 47]. Person-centered care guidelines call for engaging family and friends in shared decision-making and communication [45], and caregivers' ongoing relationships with residents' families, grounded in longstanding bonds and knowledge, position them as advocates who enable person-centered practice [8].

2.2 Family Involvement in Care

Even after a loved one is admitted to a long-term care facility, family caregiving continues [48]. However, the type of care evolves: families transition from direct physical assistance to building relationships with care staff, offering valuable insights that could greatly enhance care, and participating in care planning to discuss needs and decisions regarding a loved one's condition [24]. Yet, barriers to family-involvement remain, including strained staff-family relationships [24, 46]. Meaningful family participation is more likely when communication is regular and transparent, interactions are grounded in respect and trust, and staff explicitly acknowledge family needs and concerns [4].

Families regard communication as a means to foster collaboration between themselves and the staff, aimed at enhancing the resident's well-being [5]. However, significant barriers can still make family members hesitant to engage in a loved one's care. These barriers include insufficient support from care staff or management, strained staff-family relationships, reluctance to embrace change, and other administrative challenges [26]. Family satisfaction has been shown to depend on how much family members feel they are actively involved by the facility in delivering personalized care to their loved one [49]. Another study that measured the relationship between family involvement and family perceptions of the quality of life of care recipients in long-term care facilities demonstrated that frequent communication between families and facility staff resulted in higher family perceptions of resident quality of life. Therefore, facilities should be encouraged to become more family-oriented through encouraging communication between staff and families [45].

Family involvement is essential for delivering the best care for individuals with dementia, especially when it comes to personalized care. This personalized approach is a key element of high-quality services provided by long-term care facilities [49]. Despite this recognized significance, there is a scarcity of interventions that address the communication and relationship-building challenges faced by caregivers and family members [3]. Additionally, few studies explore the potential of technological interventions to alleviate the resource limitations experienced by staff

and to address these specific challenges. This work contributes to the literature by inquiring into the role of family–staff communication in long-term care, and it presents findings that underscore the importance of engagement between families and care providers for residents’ well-being.

2.3 Family-Centered Digital Caregiving Tools

Despite rapid growth in digital supports for caregiving [14], their potential to enable family involvement in dementia care within long-term care facilities remains underexamined. There is a gap in the literature regarding what information families want about their loved ones’ health and status, how they would like to receive these insights, and how this information would affect their involvement in the person’s care. Closing this gap is essential for designing family-centered tools that fit resource-constrained workflows rather than adding burden.

When not properly facilitated, overburdened staff can experience family interactions as an added burden, leading to avoidance, perfunctory updates, or strained relationships [55]. Nonetheless, evidence-based strategies can reliably foster constructive engagement—particularly relationship-centered practices that honor residents’ uniqueness and share concrete experiences of care [26]. Yet there remains no consensus on the content and structure of routine staff–family communication, underscoring the need for tools that make effective practices tractable in resource-constrained settings.

A systematic review found that among the 29 studies surveyed regarding interventions to foster family inclusion in nursing homes for care recipients, only two aimed to create family–staff partnerships, and only eight focused on enabling family members to make better-informed decisions and/or participate more actively in future discussions with staff [3]. These studies were limited in their scope, mostly concentrating on family members’ with regard to their decision-making in end-of-life care rather than measuring whether the interventions led to a greater perception of family inclusion in PLWD’s care. Therefore, interventions should consider a broader, more holistic, and person-centered role for family members of people with dementia in long-term care facilities. In all the surveyed literature, reviews, and meta analyses, interventions mostly consisted of staff training programs, facilitated family conferencing, compassion-based care programs, and toolkits and booklets to aid conversations. Scarcely any interventions took aim at examining how digital tools and platforms could potentially enhance staff information sharing and provide updates on patient care to families.

One promising study investigated the complex and challenging dynamics of family involvement following a care recipient’s admission into long-term care facilities through the promotion of communication opportunities via an individualized technological database. This database consisted of personal favorite music, photos from meaningful life events, and special souvenirs and other artifacts that were collected by the spouses of the care recipients. The personal material was found to encourage engagement and social interaction by assisting the spouses to find ways to communicate with their partners living with dementia, re-engage with shared past experiences, and reduce stress and feelings of disconnection during visits [17].

We seek to understand, improve, and build upon the state of communication not just among caregivers and between caregivers and care recipients but also between facility staff and family members of the care recipient. As illustrated by prior interventions, effective information sharing cultivates trust between families and care facilities, alleviating family anxiety and strengthening relationships with the care team. Going further, our research incorporates human-centered design to investigate how technology can minimize communication barriers, enhance family satisfaction with facility staff’s caregiving, and promote more person-centered care through enabling new avenues of family involvement.

3 Method

In this section, we outline our human-centered design approach, the participants and care facilities we collaborated with, and the methods and tools we used. All protocols were reviewed and approved by the Institutional Review Board (IRB) at Dartmouth College.

3.1 Design Framework

To orient our research process, we took an iterative design approach (Double Diamond) [15] through the lens of Podgorski et al.'s biopsychosocial-ecological framework. This framework emphasizes the integration of biological, psychological, social, and ecological factors in dementia care, therefore incorporating a complex understanding of the coordinated efforts of families, professional caregivers, and residents themselves [45]. This approach grounded us to better understand individuals with dementia in relation to their family dynamics and social networks, facilitating a holistic perspective on their personal and relational needs within the broader dementia care context.

In addition, we approached long-term care facilities as complex adaptive systems, or ecologies, as described in Section 2. This perspective helped us anticipate the inherent challenges involved in designing scalable, person-centered digital interventions. Specifically, we recognized that emergent behaviors among various agents—human stakeholders (e.g., caregivers, residents, families, administrators) as well as digital agents—are dynamic, nonlinear, and constantly adapting based on interactions. Under such conditions, interventions that cannot adapt to changing contexts and variable patterns of user engagement tend to lose their effectiveness [11, 12, 25, 47]. To address these concerns through our process, we took an expansive view of our user bases in our biopsychosocial-ecological strategies, engaging, to varying extents, a sample of users that covered nearly the complete care ecosystem at all stages of our process.

We employed a four-phase Double Diamond design process—involving **Discovery**, **Definition**, **Development**, and **Delivery**—to investigate, design, and evaluate our intervention aimed at enhancing family involvement in dementia care at long-term facilities and supporting staff caregiving practices. In doing so, we addressed our three research questions (RQ1, RQ2, and RQ3) at different stages. Specifically, Discovery centered on understanding the experiences and challenges of families and caregivers (RQ1); Definition and Development addressed how we could design and implement interactive features to meet the identified needs (RQ2); and Delivery focused on real-world system deployment and evaluation of its impact on family engagement and staff practices (RQ3).

Grounded in this theory, our approach seeks to create a technology that enhances family engagement through transparent, accessible communication within dementia care facilities. The following sections detail the methods used in each phase, describing participant recruitment, data collection procedures, system development, and analytic approaches.

3.2 Clinical Collaborators, Care Settings, Participants, and Recruitment

3.2.1 Collaborations with Care Providers. To ensure the development of dementia care technology that is directly grounded in real-world experiences, needs, and domain expertise, we partnered with a team of memory care providers from Memcara, which specializes in person-centered care and provides in-person and web-based services to care facilities across North America.

3.2.2 Care Settings. To identify potential care sites, we leveraged Memcara's connections to long-term care facilities. Our inclusion criteria were that facilities be resource-constrained¹.

¹Specifically, a resource-constrained long-term care facility is one in which the demand for staff time, expertise and equipment routinely exceeds the available supply (e.g., due to high resident loads, high acuity of residents, or shortages of trained staff). This creates chronic capacity strain—there are not enough personnel or resources to meet residents' needs, leading to reliance on overworked staff and a limited ability to provide optimal care [2, 23].

We relied on facility administrators to determine if the facility would be interested in participating, as well as to verify Memcara’s determination that the facility met our inclusion criteria². Ultimately, we partnered with seven care facilities located across the U.S. and Canada, which we refer to as F1–F7. As is common for long-term care facilities, none of our participating facilities specialized in dementia care, but each included residents who were older adults living with moderate to severe dementia.

3.2.3 Participants and Recruitment. Through collaboration with facility administrators at F1–F7, we recruited 40 professional caregivers (including nurses, certified nursing assistants, and therapeutic recreation specialists), 35 PLWD as care recipients, and one representative family member per care recipient, for a total of 35 family members. Following our IRB-approved enrollment procedures for a study involving older adults and PLWD, we obtained written informed consent or proxy consent where applicable. Family members were contacted via facility-wide email announcements, posted flyers, and word of mouth; staff caregivers were recruited with the support of each site’s administrative team.

3.3 Discovery (Addressing RQ1)

In the Discovery phase, we sought to answer RQ1 by exploring how families and caregivers experienced day-to-day communication, what barriers they encountered when capturing, using, and sharing information on PLWD, and what opportunities might exist for improving family involvement.

Field Notes. We carried out a year-long observational study, conducting weekly site visits across all seven locations to observe and document care practices and interactions between staff, residents, and families in real-world contexts. Observational field notes captured issues such as staff workload, technology usage, and the emotional tenor and pain-points of impromptu family visits and quarterly family-staff care conferences.

Interviews. In parallel, we conducted 16 semi-structured interviews, each lasting 60 to 90 minutes, with the subset of family members and staff caregivers. The semi-structured format allowed for flexibility, enabling participants to share in-depth personal experiences while still addressing key themes we sought to investigate. When possible, interviews were conducted in person at the facility, but some interviews were conducted over Zoom due to scheduling or location constraints. Interview questions explored how participants currently managed communication about a resident’s health and routines, the types of information families most valued, and any resource or regulatory constraints that shaped staff–family collaboration. Interviews were recorded when permission was obtained to do so.

3.4 Definition (Addressing RQ1 and RQ2)

We transcribed all interview audio recordings using software and consolidated our field notes from observations into a shared repository. Three coders each performed in vivo coding on distinct interviews to capture participants’ own language around perceived communication gaps, staff scheduling pressures, and family members’ desire for real-time updates. Our synthesis primarily occurred during weekly research team meetings, which included representatives from Memcara. During these meetings, we reviewed and refined our coding categories based on recent observations and identified emergent themes, such as how short staffing and privacy regulations limited staff–family engagement. From the refined thematic clusters, we subsequently synthesized them

²We asked about the facility’s access to critical resources (staffing, funding, equipment, training), whether the facility was struggling to meet enrollment requests and family needs, and the ratio of residents to staff.

into key insights into how information on PLWD is recorded, shared, and used to provide care and create a sense of involvement for family members. We compared our own findings to the established literature on family involvement in care, noting agreement, differences, and enrichment. To enhance clarity and coherence in this paper, we focus specifically on presenting results related to information-sharing practices. Building on the results captured from the Discovery phase, the Definition phase aimed to translate observed challenges into concrete design imperatives. This phase laid the groundwork for RQ2 by conceptualizing the types of interactive features and data-sharing mechanisms that could address the identified themes. Finally, we derived specific functional requirements for our intervention.

3.5 Development (*Addressing RQ2*)

We next developed low- and high-fidelity prototypes to explore technological features aligned with the established design imperatives. We called our system "Family In The Loop" or "FITL." The name is intended to capture the ideas of facility staff-family member collaboration as well as human-AI collaboration (in reference to the phrase "human in the loop" [40]). Complementing the methods presented here, specific user feedback and general themes presented in Section 4 will further clarify how the core challenges identified in RQ1 drove our design goals and iterative development process.

3.5.1 Initial Design Goals. Understanding that technology adoption is a challenge in this domain, we drew on Bezboruah et al.'s work which suggests involving staff in identifying existing effective strategies and then leveraging these as models in our design process [9]. Following this procedure, we created several features to address user needs. These included video, image, and audio update interfaces; text and audio chat-style Q&A for family-staff communication; and separate interface views for family members and for staff, each providing detailed care notes. See Figures 5 and 6 in the Appendix for examples of these prototypes.

3.5.2 Iterative Prototyping.

System Adjustments: To evaluate and iteratively refine our prototypes, we employed the think-aloud method during simulated caregiver scenarios [19, 37]. Initially, we explored a wide range of interfaces through low-fidelity prototypes—such as paper sketches, storyboards, and simple wireframes [52]—sharing these early designs with a subset of caregivers and family members (n=16) for feedback.

Low-Fidelity Prototypes: During scenario-based testing, we presented participants with multiple low-fidelity prototypes and a series of prompts simulating real-life care events that varied in complexity and urgency. Participants were asked to navigate the prototype based on their natural reactions to these prompts, allowing us to observe how their engagement levels fluctuated depending on the scenario. These scenarios were designed to assess how family members would prefer to receive and interact with care-related information under different circumstances. For example, one scenario depicted a routine, non-urgent update about their loved one attending a scheduled group activity, while another described a more critical situation, such as an unexpected change in medication or an incident requiring staff intervention.

High-Fidelity Prototypes: Based on insights from early prototyping, we developed and iteratively improved a single high-fidelity prototype – a mobile-first web application backed by real-time data entry via Firebase. We tested our high-fidelity prototype using the Wizard-of-Oz protocol, which involved an additional interviewer who actively updated the database in response to user interactions [16]. The web application listened for Firebase updates, creating the impression of a fully dynamic, functional system. To keep our prototyping nimble, we simulated particularly

complex UI components as static images that we could update manually and re-upload. Loading bars, which we could quickly trigger on or off, as well as an explanation that this was a new, AI system that took time to process feedback, helped make the lag in updates appear more realistic. As a temporary stand-in for the not-yet-complete backend integration with existing systems, we mocked up a small sample dataset with expected results. Before testing with caregivers at facilities, we tested with care experts from Memcara, organizing feedback as a heuristic evaluation [41]. Each walkthrough with caregivers was conducted by two interviewers who took notes and then led a discussion of the findings in weekly meetings.

Functional Prototype: Having created a high-fidelity specification for FITL that addressed the question (RQ2) of how to design an interactive technology that supports the identified needs and enhances relationship-building to foster a greater sense of engagement for families of PLWD, we finally implemented a fully functional prototype. This involved programming a robust frontend, creating backend integrations with Memcara for passive data collection, and the development of a data processing pipeline.

3.6 Delivery (Addressing RQ3)

3.6.1 Pilot Testing and Refinements. Prior to the broader deployment of the system, we conducted multiple pilot tests with small groups of staff caregivers and family members of care recipients at two facilities, which had been using the prototype for three weeks by that point. Interview debriefs, surveys, and system usage logs helped us identify strong points (e.g., families valued the emotionally resonant short videos) as well as friction points (e.g., the AI-generated bullet points needed clearer explanations and citations). We refined the back-end data pipelines for video capture and analysis, ensuring compliance with privacy guidelines by redacting unrelated or sensitive footage. Through these refinements, we arrived at our final version of FITL.

3.6.2 Deployment and Evaluation. Having completed a pilot test, we deployed FITL at partnering care facilities to evaluate FITL's real-world impacts on family engagement and staff caregiving practices. In our deployment, we were careful to mitigate additional workloads on the staff. Ultimately, we limited our deployment to four of the seven participating facilities due to unanticipated administrative departures at the excluded sites³. We provided short training sessions to each site's care staff on using FITL. Family members received secure login credentials and a video-introduction. This deployment lasted three weeks at each site, enabling us to observe whether families felt more involved in their loved ones' care and whether staff found the system's data-sharing tools efficient or burdensome. We decided on three weeks as an appropriate time frame because it allowed sufficient time for staff and family members to develop familiarity with the system, engage in multiple interactions, and provide meaningful feedback—while remaining short enough to minimize potential disruption to care routines and staff workload. To gauge FITL's impact on families' sense of involvement in care as well as other caregivers' practices (RQ3), we used on-site observations to assess how staff integrated the FITL tools into their daily workflow and the effects on care outcomes. We conducted in-depth interviews with participants who could find time outside of work hours (six family members and eight staff caregivers) at the end of the trial to understand shifts in perceptions of communication quality, family involvement, and trust in the caregiving team. Participants who were not willing to participate in interviews cited scheduling challenges as a reason for excluding themselves. Qualitative data collected during the deployment phase (interviews and on-site observations) was thematically analyzed by three coders following the same approach discussed in Section 3.4.

³After consulting with the staff in these cases, we decided to discontinue the study due to concerns that interviews and outside observers would distract the remaining staff at a difficult time.

4 Findings

We begin with insights from our Discovery phase (RQ1), which surfaced a range of interwoven, information-related challenges. We then describe how these findings informed the design and development of our prototype (RQ2), before concluding with an evaluation of the deployed system and its impact on family engagement and caregiving practices in real-world care environments (RQ3). In reporting on themes throughout this paper, we refer to staff caregivers and family members using “Fx-CGy” and “Fx-FMy”, respectively, to identify their facility ID (F) and participant ID prefixed to indicate their status as a caregiver (CG) or family member (FM).

4.1 RQ1: What are the experiences of families and professional caregivers of care recipients in long-term care facilities, particularly with respect to challenges in communication and family involvement?

We conducted a qualitative investigation into the experiences of families and professional caregivers at long-term care facilities. Two central themes emerged from our analysis: (1) **Staff-Centered Caregiving Needs**, capturing staff caregivers’ experiences in dementia management⁴, and (2) **Family Involvement & Information Needs**, highlighting how families engage and seek transparency in care.

Dividing our findings into these role-based themes allowed us to surface the distinct—but often overlapping—needs, practices, and constraints faced by each group. This structure helped clarify how breakdowns in one role’s experience often ripple into the other’s, and enabled us to better analyze tensions, expectations, and opportunities for improved coordination. We return to the question of how roles are negotiated and evolve—particularly in high-involvement cases—in Section 4.1.3.

4.1.1 Theme 1: Staff-Centered Caregiving Needs. As discussed in Section 2, dementia care adds significant complexity to already demanding caregiving work. In addition to routine responsibilities like hygiene and feeding, staff must respond to unpredictable behaviors, emotional volatility, and needs that change not just daily, but hour to hour. These interactions often depend on subtle, personal knowledge—what calms one resident might distress another—and mistakes in redirection or communication can have serious consequences. Yet this person-specific knowledge is often missing, fragmented, or inaccessible in existing systems.

Staff operate in environments with high turnover, limited continuity, and tools that rarely support the kind of flexible, context-rich care that dementia demands. To understand where current systems break down and where design can help, we organize our findings into three subthemes: **the challenges staff face** in daily dementia care, **the workarounds and informal practices** they use to fill gaps, and the **design constraints** that shape what kinds of interventions are actually feasible on the ground. These subthemes reflect how staff make care work—not just through protocols, but through improvisation, communication, and shared knowledge that often goes undocumented.

⁴Beyond what is noted in Tables 1 and 2, we initially attempted to further distinguish between general caregiving needs and those specific to dementia, but found that dementia shaped nearly every aspect of care—sometimes in subtle ways, and sometimes in more pronounced ones. In most cases, it compounded practices that were already under strain. While this paper focuses specifically on dementia care, we believe the approaches proposed here may be applicable more broadly, given that dementia represents a particularly challenging use case. We leave it to future work to further examine this distinction.

Table 1. Theme 1: Staff-Centered Caregiving Needs

Subthemes (with associated codes)	Description
Staff Challenges (rooted in the demands of dementia-specific symptoms)	
Symptom management support	Managing challenging dementia behaviors.
High workload	Heavy demands limit dementia-specific caregiving.
Staff burnout and turnover	Turnover disrupts dementia-specific caregiving.
Family-Facing Communication	Communicating what residents can no longer express.
Staff Existing Practices & Limitations (illustrations indicate ties to design decisions in Section 4.2)	
Briefings on care essentials	 Notes on managing dementia symptoms.
Internal staff conversations	 Conversations about care recipient specific info.
Archives of records	 Standardized records to meet legal requirements.
Staff Design Constraints (broadly applicable to care beyond PLWD)	
Minimal caregiving disruption	Solutions cannot increase caregiver workload.
Workflow integration	Tools must align with existing caregiving practices.

Staff Challenges.

Symptom Management Support. Staff repeatedly emphasized that effective dementia care depends on knowing each resident’s triggers and calming strategies. As F3-CG1 explained, misunderstanding a resident’s needs can escalate into false reports of aggression:

“Most people think that, as a caregiver, you want to hug them, but [this resident], she would whack you... You’ve created an incident where she looks like she’s violent because [you] are trying to redirect her in a way that is not appropriate.”

Instead, staff described how a simple memory cue—such as referencing “The Golden Girls”, which the resident watched with her sisters—could soothe her. But this kind of detail was rarely available to new or rotating staff. As F2-CG2 noted, the lack of centralized, individualized profiles meant staff had to rely on guesswork:

“If I was coming in as a new employee, I wish I could click on a name and I could see: this is their family history... these are the activities they’re successful at... these are triggers—that if they’re gonna be aggressive, don’t do these things.”

High Workload. Caregivers reported being stretched thin, with little time to reflect, observe, or document nuanced changes in behavior. As F4-CG1 noted, tools that demand real-time responses are unlikely to succeed:

“Any tool that relied on real-time responses from staff would likely fail unless it fit into moments when staff had bandwidth.”

The pace of work often meant documentation was reduced to checkboxes and surface-level entries: “cleaned,” “fed,” or “agitated,” which contributed little to ongoing care quality.

Staff Burnout and Turnover. Participants linked high turnover to a breakdown in continuity of care. When experienced staff left, their knowledge of residents—what worked, what didn’t—was rarely captured or transferred. F3-CG2 stressed that staff burnout limits caregivers’ ability to go beyond task completion:

“The system should give the caregiver a break... so they can do their own self-care and better focus on what they need to do.”

Family-Facing communication. Staff viewed family communication as essential but resource-intensive. They want to share meaningful updates—especially moments of care recipient joy and progress—but without overwhelming themselves or families. Caregivers also noted that without regular, proactive updates, families sometimes doubt that care is happening at all. For example, a family member might call and ask a care recipient what they’ve done that week, only to hear, inaccurately, that they’ve been alone or locked up—simply because the care recipient doesn’t remember. These moments can create unnecessary distress and conflict, even when care has been consistently provided.

Beyond emotional reassurance, staff saw regular updates as an important way to help families understand how their loved one is changing, and what kinds of care strategies are now needed to bring out moments of clarity, calm, or connection. These updates help families build more realistic expectations and feel more connected to the evolving care process.

Staff described a range of family communication needs, which varied in intensity and purpose. At the most basic level, caregivers emphasized the importance of lightweight updates—brief notes, photos, or short videos that offer reassurance and visibility into everyday care without placing additional burden on staff. These updates were especially helpful for sharing moments of connection or progress, and for coordinating visits based on care recipient rhythms, such as after music therapy when they were most alert. As F3-CG1 explained, “This would be the most advantageous time for visits.”

Beyond periodic updates, caregivers also noted the need for a mechanism to support clarifications or follow-up questions from families. These often arise when a care recipient provides an inaccurate account of their experience—such as saying they’ve been alone or locked up all week, simply due to memory loss. In these cases, caregivers wanted a way to share more detailed context or documentation without being drawn into ongoing back-and-forth messaging.

Finally, for families who are more actively involved—or who are concerned about consistency in care—caregivers described a need for systems that can offer accessible, on-demand evidence of care. This might include activity logs, behavioral notes, or short annotated media clips that show care patterns over time. However, staff cautioned that systems should avoid overwhelming families with raw or decontextualized data, which could misrepresent the natural fluctuations of dementia or unintentionally erode trust. Across all tiers, caregivers expressed a need for communication tools that are responsive to family needs but aligned with the realities of their workload and the nature of dementia care.

Staff Existing Practices and Limitations.

Briefings on Care Essentials. Caregivers often create informal “cheat sheets” with quick tips on managing resident behaviors. These are typically posted on walls, left in binders, or shared verbally during shift changes. However, these tools are inconsistently maintained and rarely updated, making them unreliable during staff transitions. F4-CG1 described a failed attempt to share knowledge via a physical notebook:

“Let’s say the daughter writes in the notebook every day... Then one day she forgets. The next day the housekeeper puts it in a drawer. The next thing you know, the notebook’s under clothing. The system just fails.”

Reviewing Records. Caregivers sometimes review resident charts, intake forms, or digital notes before a shift—especially when covering a new wing or unfamiliar resident. These records, often

embedded in electronic health record (EHR) systems, are meant to provide a foundational understanding of the resident's history. However, staff repeatedly told us that these systems are difficult to navigate, poorly maintained, and rarely helpful for day-to-day decision-making in dementia care.

Many caregivers described situations in which missing or buried information—such as behavioral triggers, preferred redirection strategies, or personal history—led to avoidable distress. F3-CG1 shared an example where a resident with an aversion to touch was approached by a new staff member unaware of her sensitivities, provoking a negative reaction and resulting in a mischaracterization of the resident's behavior. Without timely access to context, she explained, staff risk escalating situations that could have been prevented:

"You've created an incident where she looks like she's violent because [you] are trying to redirect her in a way that is not appropriate."

Even when resident-specific context is captured somewhere in the record, it is often buried within lengthy, compliance-driven documentation that caregivers don't have time to sort through. This mismatch between what's documented and what's actionable can have serious consequences—as seen in incidents where staff unknowingly repeated harmful actions due to missing or inaccessible information.

Beyond its importance in moments of direct care, caregivers also highlighted how personalized context supports broader responsibilities like activity planning and family engagement. F1-CG1, a programs manager, explained how reviewing resident activity logs helped her identify patterns in engagement and tailor programming more effectively:

"This person is always saying 'No'... but we have to look at why they're saying 'No'. They just like to keep to themselves? Or is it because we're not offering them anything they want to be doing?"

She noted that this type of insight is also helpful during family meetings, when staff need to explain what has been tried, what has worked, and what adjustments are being made. Yet this information is typically scattered across paper logs, mental notes, and disconnected digital systems—making it difficult to retrieve when needed most.

Staff Conversations. Caregivers rely heavily on brief verbal exchanges—often in hallways or during shift transitions—to share timely observations and coordinate care. These informal handoffs are a vital part of how teams adapt to changing resident needs in real time. However, caregivers noted that these conversations are often unstructured, easily forgotten, and highly dependent on individual memory and follow-through. Participant F4-CG1 explained how communication is often constrained

"There is not enough time to have a sit down every day... I could say to the social worker, 'I realized today that Mrs. Jones needs new clothes,' and she says, 'Okay, I'll call the daughter.'" However, if the social worker forgets, this communication also fails."

While these conversations are essential to keeping care running, staff repeatedly emphasized their limitations. Important details—both short-term updates like a clothing need or changes in mood, and longer-term insights such as a resident's response to certain activities or preferred redirection strategies—can easily slip through the cracks if not captured or reinforced elsewhere. As one caregiver noted, this kind of person-specific knowledge is especially vulnerable to loss during shift changes or when staff rotate across units.

Caregivers expressed frustration at the fragmented nature of current communication practices, with information spread across notebooks, whiteboards, casual conversations, and digital systems that rarely sync. As F1-CG1 put it:

“It’s all in different pockets right now, not all on one platform.”

Together, these reflections point to the need for tools that preserve the immediacy and informality of verbal exchange while ensuring important information is captured, retrievable, and shared across roles and time.

Staff Design Constraints.

Information Continuity. Caregivers emphasized that resident-specific insights—such as triggers, calming techniques, and care preferences—must persist across staffing changes and organizational churn. These insights are often hard-won and critical for avoiding repeated mistakes. F7-CG1 shared a tragic incident in which a care recipient, due to stroke-related language confusion, repeatedly said “yes” when she meant “no.” Without access to this crucial detail, a rotating staff member misunderstood her agreement as consent, leading to serious physical harm and legal consequences. Although this information existed in the digital record, it was not surfaced to the caregiver in a timely or accessible way.

Minimal Caregiving Disruption. Any new tool or process must respect the constraints of fast-paced care environments. Caregivers emphasized that systems requiring extra effort, multitasking, or cognitive load are unlikely to be adopted. As F2-CG1 put it, the system should “help us—not slow us down.”

Workflow Integration. Designs should extend—not replace—existing informal practices like resident cheat sheets, conversations between caregivers, and deep dives into records. Tools must adapt to the rhythms of care work and slot into existing touchpoints without demanding new ones.

Reflection on Staff’s Dementia Caregiving Needs. One of the clearest takeaways across our interviews was that care recipient-specific knowledge—such as behavioral triggers, calming strategies, and preferred routines—is essential to high-quality dementia care. Yet caregivers consistently noted that this information is fragile: it often resides in the heads of long-tenured staff, scribbled on informal notes, or buried deep in compliance-oriented records that are hard to access. When staff rotate or leave, that knowledge is frequently lost.

This fragility is not merely inconvenient—it can lead to real harm. F7-CG1 shared a tragic example of a care recipient who, due to progressing dementia symptoms, consistently said “yes” when she meant “no.” This detail had been documented, but it was never surfaced in a way that new staff could easily access or understand. As a result, a rotating caregiver misinterpreted the resident’s responses, leading to serious injury and legal consequences. Although the system had recorded this insight, it failed at the point of care.

This incident reflects a broader pattern identified in Theme 1: *caregivers are doing complex, relational, high-stakes work in systems that do not support the continuity their work demands.* These failures are compounded by high workloads, rapid staff turnover, and fragmented information-sharing practices. As described in previous sections, information gets passed in hallway conversations, written on cheat sheets, or tucked away in binders—none of which reliably persist across time or roles.

Designing for **information continuity** requires more than just building digital records. It requires creating mechanisms that surface what matters, when it matters—without burdening staff with extra documentation or workflows. As caregivers made clear throughout our interviews, the goal is not just more information, but more accessible, actionable, and enduring knowledge

that supports care over time. Supporting continuity, then, means designing within real-world constraints: tools must fit into tight workflows, preserve the nuance of person-centered care, and work across rotating and unevenly trained teams.

4.1.2 *Theme 2: Family Involvement & Information Needs.* Families remain emotionally and practically involved even after their loved ones enter residential care. This role becomes more complex as dementia progresses and residents lose the ability to self-advocate. Families depend on staff for regular updates but often feel excluded or under-informed about daily experiences.

Table 2. Theme 2: Family Involvement & Information Needs

Subthemes (with associated codes)	Description
Family Challenges (rooted in the demands of dementia-specific symptoms)	
Limited Visibility into Daily Care	Communicating what a loved one can no longer express.
Eroded Trust in Care Quality	Lacking trust in staff’s dementia caregiving.
Emotional Responsibility Without Agency	Feeling responsible but unable to influence dementia care.
Family Existing Practices & Limitations (repeated illustrations from Theme 1 indicate similarity, see Figure 2)	
Informal updates	 Occasional updates on care recipient’s status.
Conversations with staff	 Follow-up conversations about care updates.
Care conferences	 Scheduled meetings for care review and discussion.
Family feedback	 Families share stories and care tips.
Family Design Constraints (broadly applicable to care beyond PLWD)	
Family-staff coordination	Family knowledge integration cannot burden staff.
Privacy and compliance clarity	Information sharing must navigate privacy regulations.

Family Challenges.

Limited Visibility into Daily Care. In dementia care, where residents often cannot reliably self-report, families rely on staff to understand how care is unfolding. Many described frustration with inconsistent updates—typically shared through sporadic emails, calls, or in-person conversations—leaving them without a clear sense of whether care reflected their loved one’s needs. As F4-FM6 shared, “*I want to feel confident that my loved one’s care reflects her needs and preferences, even when I’m not there.*”

Families expressed a desire for structured, tailored communication: regular briefings, the ability to ask clarifying questions, and access to supporting details when needed. These moments helped them stay engaged and aligned with care goals, especially when routines or behavioral responses changed. At the same time, families recognized that time constraints, documentation challenges, and staffing shortages made consistent communication difficult. Rather than constant contact, they sought systems that supported flexible, layered information sharing without adding burden to caregivers—helping them feel informed, involved, and emotionally reassured.

Eroded Trust in Care Quality. Limited visibility often fed into a broader erosion of trust. In dementia care settings, where families cannot rely on the care recipient’s memory or verbal accounts, trust in the care team becomes especially central. Families described a lingering sense of

unease about whether care was being delivered consistently and attentively. These concerns weren't always tied to specific events but accumulated over time—through inconsistent staff interactions, lack of follow-through, or mismatches between what families observed and what they were told. As F5-FM2 noted,

“The care my mother receives has improved somewhat, but inconsistencies remain. Some staff members consistently provide excellent care, while others seem less engaged, creating significant uncertainty in the quality of care.”

Trust, families emphasized, was not solely about visibility. It was shaped by the tone and responsiveness of communication, a sense that staff understood and respected the care recipient's needs, and prior experiences of either feeling included or left out. In dementia care, these relational aspects are heightened: because care recipients may struggle to self-advocate, families must rely on staff not only for care delivery but also for interpretation and representation of their loved one's needs. Without systems that support consistent, empathetic communication, trust was easily strained.

Emotional Responsibility Without Agency. Many families described the emotional weight of feeling responsible for their loved one's wellbeing—especially in dementia care, where the care recipient may be unable to self-advocate—while lacking meaningful ways to influence care. As one family member put it,

“I still consider myself her primary caregiver... I help manage anything that needs to be managed” (F4-FM1).

Despite this sense of responsibility, families often felt excluded from decision-making. They described offering input or asking questions, only to see little follow-through and repeated incidents of avoidable caregiving errors. This disconnect left them feeling anxious and powerless, caught between wanting to help and lacking the tools or opportunities to do so.

Existing Practices and Limitations.

Informal dementia updates. Some families described receiving occasional updates via email or SMS, sometimes accompanied by photos or video clips. These moments were deeply meaningful—such as a short video of a care recipient singing along during a group activity. F5-CG1 shared an example of calling a family member after a previously nonverbal care recipient began singing along during a group activity:

“I called the family, and I just said... ‘You know that your loved one actually started singing along.’ And they were so happy and touched that I reached out to them for something like that.”

However, these kinds of updates were more aspirational than routine. Caregivers explained that recording and sharing media was difficult to scale. Capturing a moment of lucidity in dementia care often requires timing and attentiveness that conflict with the demands of active caregiving. Attempting to record during care may disrupt the moment itself, while recording continuously and reviewing footage to find shareable clips is far too time-intensive. As a result, media updates were generally left to administrative or managerial staff, and occurred infrequently. Caregivers also expressed caution about taking initiative in sharing updates due to unclear boundaries around privacy and communication protocols. This left families wishing for more regular insight, but unclear on whether or how it might realistically be provided.

Conversations with Staff. Follow-up questions or clarification requests were often addressed verbally during visits or by calling the facility, but these exchanges were unpredictable. Families

valued being able to check in and ask specific questions about observed changes or strategies in use, yet noted that the effectiveness of these conversations hinged on who was available and whether the staff member had relevant knowledge to share.

Scheduled Care Conferences. Most families participated in periodic care meetings (often quarterly) where staff shared updates on the care recipient's condition, medications, and long-term care planning. One key strength of these conferences was that staff prepared in advance by reviewing and organizing more detailed records, which meant families could access summaries or explanations that were otherwise difficult to obtain. Families appreciated having this structured opportunity to ask questions and hear directly from multiple care team members. However, because these insights were not available on an ongoing basis, families still felt disconnected between meetings—especially when trying to understand day-to-day care dynamics.

Family Design Constraints.

Family–Staff Coordination. Designs should enable families to contribute knowledge and stay informed in ways that are lightweight, asynchronous, and minimally disruptive to staff workflows. Tools that require extensive time or attention from caregivers are unlikely to be used consistently and may add to existing burdens.

Privacy and Compliance Clarity. Communication systems must clearly embed privacy and compliance guidelines, making expectations transparent for both staff and families. Doing so can reduce staff hesitation around information sharing and help families understand what types of updates are appropriate and available.

4.1.3 Integrative Reflections: Entangled Roles and Informational Needs. Dementia care amplifies general caregiving challenges by introducing deep person-dependence, emotional volatility, and informational asymmetry. While staff attempt to manage this through informal workarounds, and families attempt to stay connected via ad hoc inquiries, the absence of structured, persistent information-sharing mechanisms produces gaps.

Both staff and families expressed a desire for shared, contextual knowledge that is easy to access and update. A family member put it plainly: “I want to feel confident that my loved one’s care reflects her needs and preferences, even when I’m not there” (F4-FM6). To meet that goal, dementia care systems must prioritize continuity, transparency, and communication—without placing additional strain on already overburdened caregivers.

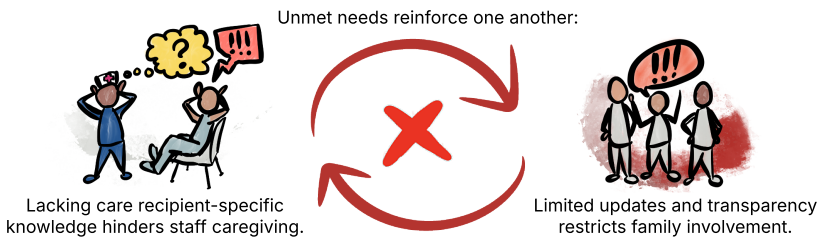


Fig. 1. Staff and family needs are deeply intertwined—when one side’s needs go unmet, it often increases pressure on the other. *All illustrations in this paper are the first author’s hand-drawn interpretations of our findings.*

Our analysis of staff and family roles highlights a key tension: although their responsibilities and proximity to care differ, their informational needs often converge. This is especially evident when family members become more involved—due to a change in condition, proximity, or advocacy role—and begin seeking not just updates, but contextualized knowledge that helps them interpret what is happening and respond accordingly.

While we began this analysis expecting to delineate staff and family needs as distinct, we found that their roles are more entangled than discrete. Staff may be oriented around documentation, compliance, and clinical decision-making, but they also serve as relational intermediaries for families. Likewise, even families who are not physically present still act as caregivers—albeit asynchronously—by contributing context, asking for clarification, and offering insights into the resident’s history and preferences.

This convergence does not mean their needs are the same, but rather that they intersect in ways that call for shared tools, not siloed ones. The challenge is to accommodate different forms of presence, authority, and emotional investment without forcing alignment. Families and staff do not simply need access to the same data—they need tools that reflect the different ways they make sense of and act on that data.

This finding guided our system design toward supporting both differentiation and overlap: tools that provide shared visibility while respecting role-based boundaries, and that accommodate the fluid, often shifting nature of caregiving responsibilities over time.

Zooming out, we also came to understand these needs as part of a broader care ecosystem—one that extends beyond the facility to include hospitals, outside clinicians, and administrative partners. Within this larger system, staff and family needs are not isolated; as shown in Figure 1, they are tightly interwoven. Breakdowns in communication or coordination ripple outward. When staff lack insight into a resident’s personal history, preferences, or behavior patterns, they may struggle to tailor care in ways that feel meaningful. In turn, families who perceive care as impersonal or inconsistent often escalate concerns, leading to repeated requests or emotional tension—further increasing pressure on already overstretched staff.

This creates a cycle where everyone is trying to do their best, but in the absence of the right information, they end up working at cross-purposes. We see an opportunity to break this cycle through what we call care transparency—the consistent, two-way sharing of contextual and behavioral insights between staff and families. In dementia care, where residents often cannot advocate for themselves, this shared information becomes the foundation for personalized, responsive treatment.

Through bringing families into an **information-sharing loop** within the broader care ecosystem, staff could access valuable context that supports more personalized care, while families would gain clarity and confidence in how care decisions are made. Designing for this kind of transparency is less about monitoring and more about building shared understanding—helping both sides align around the unique needs of the person at the center.

4.2 RQ2: How might we design and build an interactive tool that addresses the needs of family members and caregivers and enhances relationship-building to foster a greater sense of engagement for families of PLWD?

We designed FITL as a single, unified system that supports both caregiving staff and family members of PLWD. While our RQ1 analysis surfaced distinct needs and constraints for each group, we made a deliberate decision to avoid bifurcating the system into separate tools. Fragmentation of this kind is not only technically burdensome but structurally misaligned with the nature of dementia care, where family and staff needs are deeply interdependent and unfold within a shared informational and relational ecology.

Instead, we developed a shared infrastructure that offers differentiated access and interaction modes, while preserving informational continuity and responsiveness across both roles. By designing for overlap rather than separation, we were able to create a system that works within the real constraints of care environments—high workload, legal regulation, interpersonal dynamics—while enabling new modes of coordination that extend beyond what either group can accomplish independently.

The following two tables outline the specific design decisions we made in response to the themes that emerged in RQ1. In each case, we sought to emulate existing, situated practices already in use by staff or family members, while addressing their well-documented breakdowns: issues of scalability, information persistence, interpretability, and coordination friction. These design decisions are not speculative proposals, but grounded responses to the informal scaffolding and infrastructural gaps described by our participants. Each decision is shown alongside icons that reference earlier practices documented in our thematic analysis (see Section 4.1), emphasizing the continuity between what caregivers already do and what FITL aims to support and scale.

4.2.1 Staff-Focused Needs and Design Decisions. Informed by recurring staff challenges around **Symptom management support, High workload, Staff burnout and turnover, and Family-facing communication**, we prioritized continuity, low-burden interaction, and integration with existing workflows. Our design intentionally builds on staff routines—like shift notes, informal knowledge handoffs, and basic documentation—while working around the limitations that often render these practices fragile, inaccessible, or unsustainable over time.

These design decisions collectively reflect a strategy of infrastructuring [6, 30]: rather than introducing new burdens or abstracted features, we focus on extending and stabilizing the practices staff already rely on. The FITL system seeks to scale knowledge that is otherwise ephemeral and highly individualized—like what soothes a resident or when they are most lucid—into a shared, persistent record that respects caregivers’ time and attention.

4.2.2 Family-focused Needs and Design Decisions. For families, the key challenges included **Limited Visibility into Daily Care, Eroded Trust in Care Quality, and Emotional Responsibility Without Agency**. Our approach again centered on practices already in use: ad hoc phone calls, conversations with staff, and care conferences. FITL treats these practices as infrastructural elements—carrying both relational and informational weight—but supplements them with mechanisms that offer greater regularity, inspectability, and persistence without increasing the staff workload.

These design decisions build on existing human infrastructure rather than replacing it. Instead of assuming that family needs can be met through generalized portals or static dashboards, we intentionally designed for layered visibility and emotional resonance, embedding collaboration into the grain of everyday care practices.

Table 3. Staff-Facing Design Decisions (Referencing Existing Practices)

Subtheme from RQ1	Design Decision
Staff challenges	
Symptom management support	 Provide structured “Essentials” panels modeled after informal shift notes and hallway exchanges, surfacing person-specific tips often buried in records or lost during turnover.
High workload	 Introduce passive video capture to scale documentation practices without requiring staff to interrupt or alter care routines.
Staff burnout and turnover	 Ensure integration with EHRs to support continuity, drawing on existing archiving practices while reducing friction and duplication.
Family-facing communication	 Create a family-facing Q&A type interface for answering questions and getting feedback, with admin controls for content moderation.
Staff Design Constraints	
Minimal caregiving disruption	 Combine video-based capture with system integration to maintain current task rhythms and avoid parallel documentation burdens.
Workflow integration	 Align the system with existing cognitive artifacts and communication patterns (cheat sheets, informal briefings), allowing it to fold into daily work with minimal adjustment.

Together, Tables 3 and 4 show how we translated the complex, hands-on realities of caregiving into a system designed to scale without losing sensitivity to those conditions. These design decisions formed the foundation of our system prototype. In response to the fragmented, resource-constrained nature of long-term care settings identified in RQ1, we implemented FITL as a web-based application. This decision was grounded in the need for a platform that could support multi-stakeholder engagement asynchronously, without increasing the complexity or workload for care staff. This approach aligns with prior literature indicating that digital technologies can enhance the efficiency and effectiveness of care coordination by improving information generation, updating, transmission, and exchange along a care recipient’s care pathway [13]. A web app also allowed us to embed features into an accessible, low-footprint infrastructure—enabling integration with existing documentation practices while offering families continuous visibility without requiring real-time interactions.

Table 4. Family-Facing Design Decisions (Referencing Existing Practices)

Subtheme from RQ1	Design Decision
Family Challenges	
Limited Visibility into Daily Care	 Deliver short, periodic care updates with optional drill-downs, modeled after informal updates and check-ins that families found reassuring but inconsistent.
Eroded Trust in Care Quality	 Provide verifiable notes and annotated video to lend transparency to care practices, referencing the structure of care conferences but accessible asynchronously.
Emotional Responsibility Without Agency	 Enable family-submitted tips and reflections to persist across shifts, formalizing feedback loops that previously disappeared or went unacknowledged.
Family Design Constraints	
Family-staff coordination	 Use a shared interface that recognizes families as care collaborators, coordinating input and visibility through clear role-based permissions.
Privacy and compliance clarity	 Provide transparent, staff-friendly controls over what information is shared and with whom, keeping legal and regulatory concerns manageable.

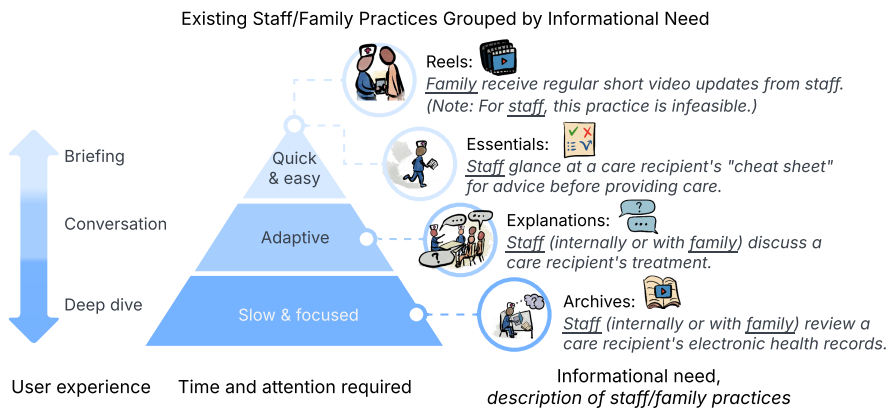


Fig. 2. Despite distinct roles, family and staff share overlapping information practices. By organizing these practices—including the aspirational model of "Reels"—around the types of informational needs they address, we highlight common ground across roles—revealing opportunities to design shared tools that support both perspectives while respecting their unique contexts.

Our iterative prototyping process focused on designing four linked system views—*Reels*, *Essentials*, *Explanations*, and *Archives*—each corresponding to a common communication practice in RQ1. These views were not designed in isolation but instead reflect an observed spectrum of informational tailoring across staff and family workflows. As shown in Figure 2, these views align with varying

degrees of time and cognitive effort required from users: Reels and Essentials serve as quick, low-effort entry points for families and caregivers, respectively, while Explanations and Archives support deeper engagement through conversation and review. In the following section, we describe how we developed and refined these components through iterative prototyping and validation with stakeholders—exploring alternative design directions, incorporating feedback across multiple levels of fidelity, and adapting the system to reflect on-the-ground constraints and preferences.



4.2.3 Backend Infrastructure: The CareInsights APIs. To put FITL into practice, we de-

veloped *CareInsights*, an interface-agnostic AI service layer with APIs for aggregating, interpreting, and synthesizing heterogeneous care data. CareInsights was intentionally designed as a standalone service with broad reusability in mind. Its development and evaluation are described in detail in a separate paper [38]; here, we focus on its role within the FITL system.

CareInsights ingests and organizes data from two primary sources: (1) video recordings of care interactions, and (2) timestamped textual documentation, including shift notes, EHR exports, and informal logs. These inputs are consolidated into a structured, queryable data layer anchored to each care recipient. Video streams are processed using computer vision and speech-to-text pipelines to produce AI-generated annotations approximating caregiver-like assessments of salient moments—such as memory recall, emotional responses, and deviations from baseline behavior. These annotations serve both as inputs for synthesis and as a direct layer of output: underlying assessments (e.g., of memory recall) can be aligned with transcript timestamps to generate overlays or timelines of interaction patterns, as later illustrated in the *Archives View* (Section 8.1.5).

On the retrieval side, CareInsights supports templated, prompt-based queries that return content in formats suited to different stakeholders or interface contexts. A lightweight, LLM-based synthesis layer filters and combines relevant records, producing responses with citations to underlying data entries—such as text excerpts or timestamped video segments. These outputs can be adapted into structured summaries, narrative explanations, or highlight-style compilations.

The backend also accepts user-provided feedback and additions—such as caregiver notes, family observations, or corrections—which are appended to the data layer and dynamically inform future queries. Because the system supports reactive querying, any previously issued or actively “listening” queries are automatically re-evaluated when the underlying data changes. This allows updates—whether prompted by error correction, new observations, or evolving context—to propagate in real time across all relevant views and interfaces using the same data stream.

FITL uses CareInsights across the four primary views, but the CareInsights APIs are interface-agnostic. By cleanly separating UI code from AI/data services, we could prototype diverse front-end representations without modifying core logic. This separation also positions CareInsights as a reusable backend for use cases beyond FITL.

4.2.4 Prototyping Process and Insights. Having established the backend infrastructure, we turned to iterative prototyping to explore how CareInsights could support and extend the care practices identified in our initial needfinding. Our process included low- and medium-fidelity walkthroughs, scenario-based usability testing, and functional prototypes populated with real-time data from the CareInsights pipeline. These iterations allowed us to refine not only the interface, but also how backend-generated content was interpreted by users in context.

Early prototypes used mocked summaries and static placeholders to explore layout, tone, and content density. As the backend matured, we began connecting real data pipelines to functional UI components. One such feature was the *Essentials* view—a concise, resident-specific summary of behavioral strategies and symptom patterns. When powered by CareInsights, these summaries were well-received by caregivers, who described the content as both accurate and immediately useful, particularly during shift transitions. The summaries reflected patterns of agitation, mood changes, and redirection strategies that closely matched staff observations—suggesting the backend could serve as a reliable cognitive support in practice.

Similarly, family-facing features such as the *Reels* view—a sequence of curated video clips accompanied by generated captions—elicited strong emotional responses. Family members found these compilations both reassuring and informative, often perceiving them as more credible and expressive than written updates. These findings affirmed that the backend could generate content resonating with diverse user needs, from emotional grounding to clinical handoff support.

However, as prototypes became increasingly realistic, new concerns emerged—particularly regarding caregiver comfort with family-facing transparency. Although many staff members initially supported the idea of sharing curated clips with families, several expressed hesitation once the stakes of real deployment became clearer. Legal liability, interpretive ambiguity, and perceived loss of control all became salient, especially when AI-selected video segments were involved. These concerns, absent during early ideation and low-fidelity feedback, underscored the gap between hypothetical and operational acceptance.

To address these concerns, we introduced a series of design controls: media content defaulted to “private,” administrative roles were given fine-grained permission settings, and a geofencing option was added to restrict access to certain content unless the viewer was physically on-site. These controls, configurable by administrator-level staff, were developed in close partnership with site administrators.

These adjustments reflect not only a necessary response to staff concerns around legal risk and content control, but also the value of sustained engagement beyond early design phases. Without long-term input, we might have relied solely on initial enthusiasm—overlooking emergent concerns that surfaced only as the system approached real-world deployment, and risking the creation of a tool staff would ultimately reject.

Another insight shaping the final system design concerned system visibility during routine care. Caregivers valued the **Essentials** summaries in theory but often forgot to consult them during high-stress moments when symptom management was most critical. To address this, we deployed QR-coded signs in resident rooms that, when scanned, opened a privacy-respecting Essentials view tied to the resident’s ID. This light-touch intervention created a physical anchor for system use, integrating the tool into existing visual rhythms and care routines.

These adjustments reflect the importance of long-term iteration and deep engagement in care settings—where systems must not only function technically but also align with shifting relational and organizational dynamics. In the following subsections, we describe the resulting views of the deployed system and the care practices it supported.

Table 5. Final Pre-Deployment Design Adjustments from Late-Stage Prototyping

Code (from RQ1)	Updated Insight	Final Design Decision
Family-facing communication	Legal concerns began to outweigh earlier enthusiasm when caregivers realized families would view selected clips.	 Added geofencing and default-private modes for staff-facing clips; gave admins granular control over content visibility.
Workflow integration	Caregivers often forgot to use the system during high-stress moments when it would be most valuable.	 Deployed QR-coded signs in resident rooms to trigger context-specific <i>Essentials</i> view; provided quick access without login.

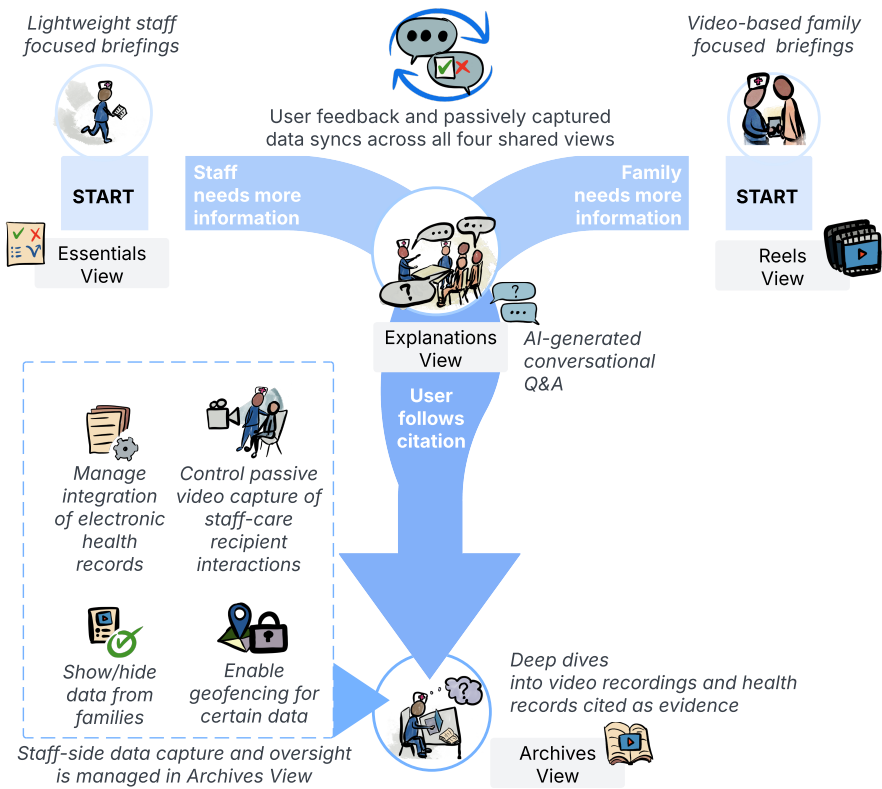


Fig. 3. Typical user journeys across FITL’s four views—Essentials, Reels, Explanations, and Archives. The flow between views reflects how users navigate escalating informational needs in response to emerging care situations, such as changes in a care recipient’s dementia-related behaviors. Feedback can be contributed at any stage and is continuously integrated into the shared data layer, enabling all views to dynamically reflect updated understanding, corrections, and newly observed details.

4.2.5 Overview of FITL System Views. FITL supports family-staff collaboration through four interconnected views—**Reels**, **Essentials**, **Explanations**, and **Archives**—each designed around existing successful practices that we observed.

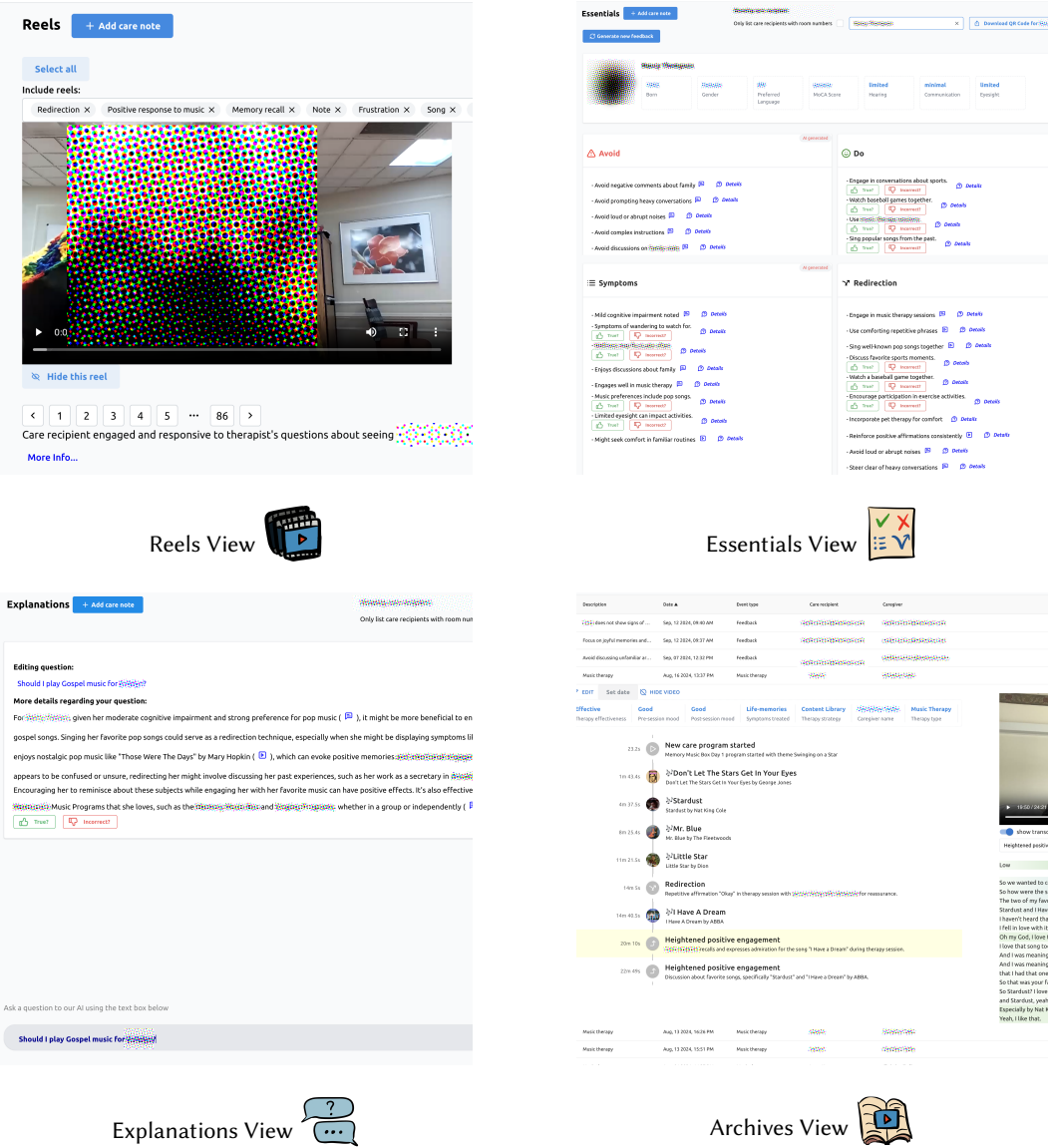


Fig. 4. Excerpts from screenshots of FITL Views, see Appendix 8.1 for a detailed overview.

Family members commonly began their interaction with FITL in Reels View, where they could see regular updates (on a single care recipient, i.e., their loved one) in the form of emotionally salient clips that offered a window into recent care. Staff, by contrast, often started with Essentials View to review tailored tips (for any of the participating care recipients) for managing dementia symptoms and supporting resident engagement. These interactions could escalate to Explanations View, which supported clarification and follow-up questions, and further to Archives View, where users could examine the records cited as evidence by the system.

Data flows into FITL through passive video capture, active user feedback, and automated integration with existing EHRs (see Section 4.2.3). This source material is used to dynamically update the AI-generated content across Essentials, Explanations, and Reels in real time when needed (e.g., in response to user feedback or a new video recording). The following sections provide a brief overview of each view; **for more detail, see Appendix 8.1.**

- **Reels View** offers emotionally resonant video updates, primarily for families, capturing moments from structured therapy sessions with minimal caregiver effort. These clips are algorithmically selected and summarized to foreground narrative or emotional meaning. They help families stay connected and offer a low-friction way to observe care that would otherwise remain invisible.
- **Essentials View** distills personalized care strategies into a concise set of bullet points, modeled after the “cheat sheets” staff sometimes informally maintain. It provides caregivers and families a shared reference for key do’s and don’ts, person-specific symptoms, and personalized redirection strategies—each grounded in system-synthesized observations that presented as citations.
- **Explanations View** emulates conversational Q&A with a knowledgeable staff member, offering narrative interpretations in response to open-ended user questions. We observed that these were typically questions about care moments—e.g., behavior changes, effective interventions-, trends, or care recipients’ memories. Like Essentials View, Explanations includes citations that a user can follow to Archives View, for instance, to index directly into the exact moment in a longer video recording where the referenced event occurred. Users can provide feedback in the form of open-ended “Care Notes” or quick, “thumbs-up/thumbs-down” interactions that, like all feedback in FITL, cascade throughout the system
- **Archives View** serves as the full-fidelity interface, enabling users to verify source records, navigate transcripts or higher-level timelines of staff-care recipient interactions, and edit the underlying content. Archives View anchors FITL’s interpretive logic and ensures transparency in how summaries, recommendations, and care notes are generated.

Together, these views tailor underlying records to the informational needs of different users.

4.3 RQ3: How does the tool impact families’ sense of involvement in care as well as other caregivers’ practices?

Having discussed user needs and the design rationale behind FITL’s features, we now turn to its impact during deployment. This section reports how the tool shaped family involvement and staff practices at four long-term care facilities over a three-week period. These findings are closely tied to those from RQ1, which documented persistent challenges in dementia care, including high staff workload, limited visibility into daily care for families, fragmented information-sharing practices, and fragile continuity of resident-specific knowledge.

4.3.1 Addressing Staff Needs.

Integration With Existing Workflows and Perceived Feasibility. Staff emphasized that FITL’s integration into daily routines—particularly through its use of existing data sources—was central to its usefulness. Several noted that the tool was available without requiring them to change how they worked. As described in the previous section, staff could engage with the tool as little or as much as they wanted, but appreciated having access to information when needed. F7-CG1 explained that FITL helped him:

“remember it and pass it on instantly while doing other things... thus saving the time of having a 15-minute conversation about each person.”

Others noted that it was especially helpful during handoffs or when interacting with unfamiliar residents, providing quick context that otherwise would have required time-consuming explanations.

The system's unobtrusive design and flexibility made it feel feasible for everyday use, especially in resource-constrained settings. At the same time, however, these same qualities—its accessibility, persistence, and integration into existing workflows—also raised concerns. As described in Section 4.3.3, caregivers and administrators expressed unease about privacy, documentation visibility, and institutional risk. In this way, the aspects of FITL that made it practical and sustainable were also linked to ongoing tensions around data sharing and governance.

Supporting Dementia Caregiving. Beyond feasibility, caregivers described FITL as a meaningful aid in the complex work of dementia caregiving. The system helped them avoid disruptions in care that stemmed from missing context. For example, F1-CG11 recounted a time when a resident became aggressive because staff unknowingly raised a sensitive topic, while F1-CG12 described a resident becoming distressed when someone else uncovered his food. In her view, FITL,

“helps prevent some of those accidental slip-ups.”

These examples suggest that caregivers saw the tool not as an added burden but as a means to reduce friction and improve care continuity.

By surfacing resident-specific context and preserving institutional memory across shifts and staff turnover, FITL supported the relational, individualized care that dementia requires. Its ability to quietly fill gaps in context—without demanding new workflows—was repeatedly cited as critical to supporting both residents' well-being and staff effectiveness.

4.3.2 Addressing Family Needs.

Streamlining Communication. Through our observations and interviews, we found that both family members and care staff felt that FITL enabled better communication through automation features, the platform's data aggregation and real-time capabilities, and search features. Users particularly appreciated how information was available at different levels of granularity, with deep links in generated summaries allowing them to drill down (switching between views) for more detail on demand. For instance, F3-CG8 expressed, *“Yeah, I think the bullet points are probably enough most of the time, but especially when it comes to redirection, the details are good to have there.”* **Family members expressed that this granular view allowed them to witness the nuanced care their loved one received, fostering a greater emotional connection and trust in the caregiving team.** Users also appreciated where FITL flags AI-generated content with a tag (see “AI generated” in Figures 8 and 9), distinguishing it from manually entered data by caregivers or family members. F9-FM1 noted that we could go even further with this feature,

“I think it'd be cool to say like, yeah, well, like 7 people found this to be true.”

Moments That Matter: Capturing Lucidity and Connection Through Reels. During deployment, Reels View offered families a way to engage with aspects of the resident experience that are typically obscured by the nature of dementia. One illustrative case involved F1-CR9, a resident with advanced cognitive impairment. In most recordings, she appeared catatonic or distressed, and care staff had been hesitant to share such footage with family members, aware that these images could be upsetting. Family members, in turn, had expressed reluctance to view recordings that might reinforce the emotional toll of witnessing a loved one's decline.

However, one clip automatically flagged by the system captured a rare moment of clarity. Toward the end of a therapy session, F1-CR9 turned to the therapist and said,

“This [therapy] gives me a reason to live.”

The statement, brief but lucid, was highly unexpected given her usual presentation. For both the therapist and her family, this reel carried immense emotional weight: it affirmed that the resident could still engage meaningfully and that the therapeutic activities were, at least in this moment, deeply felt.

This reel became especially significant when F1-CR9 passed away shortly after deployment. It remained one of the few captured moments that showed her expressing appreciation or emotion verbally, and it was shared with her family as a form of closure. Staff reflected that this type of insight would have been unlikely to surface without automated detection. Much of the footage collected during the session consisted of silence, minimal movement, or signs of distress—none of which families were inclined to view. In contrast, the reel extracted by FITL allowed families to witness a final moment of lucidity and connection. In this way, the system facilitated an experience that was both emotionally resonant and contextually appropriate, reinforcing the value of AI-augmented visibility into care recipients' inner lives.

In cases like this, surfacing a moment of deep personal meaning often also helped to strengthen families' trust in both the care team and the therapeutic process—cultivating a sense of **solidarity rather than scrutiny**, and affirming that meaningful care was taking place even when it was not immediately visible.

Supporting Clear and Actionable Recommendations for Personalized Care. Users found utility in FITL's focus on residents' triggers and preferences, with staff commenting that the available information allowed them to tailor their approach and avoid potential conflicts. For instance, F3-CG5 emphasized the importance of knowing residents' preferences, especially regarding music and conversation topics, stating that for some people it is otherwise difficult to know which memories can be discussed. FITL helped her:

"avoid controversial subjects that trigger past [negative] experiences."

These adjustments in approach, tailored to individual preferences, were described by family members as fostering a deeper empathy and connection between caregivers and care recipients. This aligns with RQ1 findings showing that emotional investment from families often drives a desire for more actionable and personalized information. A common theme was caregivers reporting that the system offered guidance on interacting with residents—what to avoid and how to redirect effectively. Family members felt this level of personalized care reinforced their involvement, especially via Essentials and Explanations Views. For instance, F9-CG1 noted:

"I'm going to look at it in a glance, in the glance, I'll read the avoid section and then I'll read the redirectable section."

Similarly, F3-CG7 praised the efficiency of Essentials View:

"It comes in handy for people that don't want to sit there and go through everything."

Finally, users appreciated how insights in FITL combined personal narratives with caregiving strategies from nursing literature. F3-CG9 remarked that FITL helped them,

"get a better snapshot of the patient, not just the medical side,"

reinforcing holistic involvement.

FITL as a Learning Tool. Many family members emphasized FITL's value as an educational tool—for themselves and for staff. For example, F5-FM1 noted that misunderstandings stemming from staff inexperience, particularly with comorbid conditions like dementia and aphasia, could undermine care. FITL helped bridge that gap. Family members described watching recorded sessions in Reels and Archives Views as opportunities to learn what strategies worked well. They appreciated how clips could be filtered and annotated with system-generated cues about resident response,

such as memory recall or emotional expression. F4-FM1 explained that these features helped her understand her mother's progression despite not being physically present. **These tools enabled emotional connection and situated learning by observing effective caregiving moments.**

Integration with Existing Systems. Throughout the study, users expected the platform to provide useful, integrated information immediately. FITL's authentication and use of Memcara's infrastructure supported this expectation. F1-FM1 appreciated that FITL consolidated care insights that would otherwise remain informal or undocumented:

"We trade information all the time... but it's kind of an assumed thing... it's not written down anywhere."

By Improving Continuity, FITL Strengthened Staff-family Coordination. Our late-stage prototyping revealed that synchronizing event logs from Memcara's interface with video allowed for precise understanding of what stimuli triggered observed responses—surfacing deeper context without increasing staff documentation workload. F4-FM1 proposed additional integrations like a split-screen view to watch stimulus and response together, likening it to "reaction videos" on YouTube. **This reflects a strong desire from family members to feel more immersed in their loved ones' daily care routines, even from afar.**

4.3.3 Ethical, Legal, and Privacy Concerns.

Privacy Protocols Hindered System Deployment. While FITL was well received, privacy concerns—particularly around video sharing—emerged as a barrier. Family members expressed frustration when limited access prevented them from connecting with their loved one's care. **Though reassured that data was protected, family members often found the administrative process for account access confusing or unresponsive.** We found that family involvement was often curtailed not by disinterest but by institutional hurdles—highlighting the tension between privacy safeguards and inclusive care.

Access to Sensitive Health Records. As FITL introduced new forms of visibility into care delivery—through video, structured summaries, and contextual records—participants surfaced a range of concerns that reflected not outright resistance, but engagement with the tradeoffs involved. While staff generally appreciated that the tool aligned with their workflows and could be used flexibly, they also pointed to challenges around visibility, including concerns about overexposure, documentation boundaries, and the emotional weight of reviewing care data.

Concerns also surfaced around access to trauma-related records or footage with other residents in view, an issue that we identified but had not adequately accounted for in our design. **Staff requested more control over what family members could access to avoid unintentional disclosure.** These features attempted to strike a balance between transparency and professional discretion. Yet even as FITL made care more visible, staff sometimes felt emotionally taxed by the idea that every interaction could become reviewable.

Family members, likewise, expressed ambivalence. They valued the opportunity to remain informed, but were often unsure about how much was being shared, who controlled access, and whether what they saw would feel supportive or distressing. Across both groups, concerns around privacy, emotional complexity, and system access highlighted the limitations of existing policies and infrastructure. Ultimately, these responses suggest that efforts to increase transparency in dementia care must be carefully balanced with emotional, relational, and institutional realities, and that even well-aligned tools like FITL can surface new tensions when deployed in complex care settings.

Legal Risks in Recording. Some users worried about documentation being misused, but most accepted that video documentation, with appropriate guardrails, could enhance care. **Family members saw these recordings as a valuable source of connection.** We also observed cases where staff were initially supportive of sharing clips but grew hesitant as deployment neared—underscoring the importance of long-term engagement to surface operational, not just hypothetical, concerns.

Families' Complex Relationship Between Privacy Concerns and Benefits. Family members overwhelmingly valued visibility, even when conflicted about data sharing. **Their concerns were not about surveillance but about maintaining their loved one's dignity and control.** For instance, F4-FM6 described the emotional difficulty of “watching from a distance” and wanting to help. As we identified in RQ1, these tensions highlight that involvement is not only a matter of access but of identity and advocacy—family members often saw themselves as co-caregivers, despite their distance.

Consent Before Data Collection. Finally, users stressed the importance of meaningful, not just legalistic, consent. **Family members wanted clearer ways to engage in decisions about data use and sharing.** F1-FM9 noted:

"I think it's important that we have a say in what information is shared and with whom."

Staff also expressed that clearer consent processes could help them manage expectations and reduce legal risk, especially as AI-generated records became more prominent in care conversations.

4.3.4 Takeaways. Our deployment of FITL across four long-term care facilities revealed both the value and complexity of integrating AI-augmented tools into dementia caregiving. FITL improved families' sense of connection and trust by surfacing emotionally resonant moments and enabling greater visibility into everyday care—often revealing aspects of resident experience that would otherwise remain obscured.

These positive experiences occurred in a care context where new technologies are often rejected if they add even minimal friction. That FITL was not only tolerated but actively used speaks to the urgency and depth of the underlying needs it addressed. Caregivers, who typically operate under intense workload pressures and with significant legal constraints around documentation, chose to adopt the system and allowed data recording that would normally raise red flags. This engagement alone is a meaningful outcome: in environments where emotional and cognitive overload are common, any tool that survives routine use signals feasibility, trust, and unmet need.

Participants voiced important concerns around privacy, emotional labor, and system boundaries—but these concerns were raised not to reject the system, but to make it work. Despite reservations, caregivers and families alike expressed a strong desire to continue using FITL. This willingness to persist reflected a shared recognition that FITL provided a form of visibility and continuity not otherwise available through existing workflows.

Together, these findings point to the need for privacy and access models that reflect the lived constraints of caregiving environments. Participants were not opposed to visibility, but emphasized that it must be implemented with attention to clinical roles, family expectations, and the emotional weight of encountering care data. Designing for privacy in this context involves more than technical safeguards—it requires supporting the relational and operational infrastructures that make data meaningful and manageable.

In the following discussion, we consider how these findings extend current understandings of family-centered care and CSCW system design, highlighting how technologies like FITL can reinforce person-centered practice while navigating the emotional, institutional, and relational dynamics of long-term care.

5 Discussion

This paper introduces **Family In The Loop (FITL)**, a digital platform that augments dementia caregivers' ability to provide person-centered, family-framed care by supporting knowledge synthesis and communication between staff caregivers internally and with family members. In our approach, we leverage prior literature on family involvement and our own discovery and definition process to deeply understand the complex needs of our different users. A central insight from RQ1 was that families were expected to trust care processes they could not see, influence, or fully understand—despite possessing critical knowledge that could enhance resident wellbeing. As staff faced systemic constraints, family insights were often lost in the churn of shift work, inconsistent documentation, and reactive caregiving. These breakdowns disrupted the continuity of care and undermined the mutual trust necessary for person-centered practice. FITL was designed to respond to this challenge by making everyday care more visible. A continuous stream of captured therapy videos, along with integrated EHR data, served as passive inputs that allowed the system to surface meaningful moments and improve its insights over time—even without direct user input. This sustained visibility encouraged families and staff to return, reflect, and learn not only from what the system highlighted, but from one another, creating a feedback loop of shared understanding and asynchronous support.

Our deployment findings (RQ3) show that FITL was adopted because it aligned with existing workflows and addressed a gap that staff and families both saw as urgent. Staff used the system despite concerns about increased visibility, recognizing that it helped them deliver care more effectively. Families felt more connected and reassured when they had access to curated updates, rather than having to seek out or interpret fragmented information on their own.

5.1 Envisioning a Data-driven, Person-centered Care Ecosystem

Stepping back, we position this paper as a step toward a future in which data-driven, person-centered dementia care is routine. In that future, AI-enabled tools and adaptive platforms synthesize longitudinal, multimodal data to reduce routine burdens and surface targeted, context-aware care suggestions. Achieving this requires moving from today's data silos and fragmented practices to collaborative, transparent family–staff interactions supported by adaptive tooling. As mediators, digital agents can streamline coordination, open dependable communication channels, and strengthen trust, helping families stay engaged and informed. Integrated in this way, AI can support caregivers in acting on a rich, shared understanding of each care recipient's needs, history, and evolving care plans. The result is not only greater efficiency and improved outcomes for PLWD, but also healthier family–staff relationships and a more sustainable, rewarding caregiving experience.

While FITL improved visibility into the daily experiences of individual residents, our deployment also revealed its limits. The system helped staff and families stay aligned, surfacing meaningful insights in the flow of care—but person-centered care does not begin or end with documentation. Caregivers were often eager to provide more individualized support, yet they worked within systems that prioritized efficiency and risk avoidance. Overmedication remained common, and time for tailored engagement was limited. Even the most motivated caregiver could only do so much. At a certain point, person-centered care becomes a matter of resource allocation: deciding which residents receive which forms of support, and when. These are not just personal or clinical choices—they are shaped by the structural design of long-term care.

Administrators and program coordinators, who manage staffing, programming, and regulatory compliance, were not the intended users of FITL. And while the system offered rich insights into individual needs, it lacked a mechanism for translating those insights into action at the institutional level. Supporting person-centered care at scale requires more than visibility. It demands tools that can synthesize patterns across residents and help decision-makers weigh those needs against legal,

financial, and operational constraints. FITL was not designed for that role—but its limitations help clarify what kinds of systems are needed next.

To meet the kinds of institutional needs that tools like FITL cannot address, we outline a conceptual architecture for a **data-driven, person-centered care ecosystem**. This ecosystem moves beyond supporting individual care decisions to helping facilities plan, allocate resources, and advocate for care practices at scale. Rather than relying on a single system, it brings together multiple layers of technology and insight, each serving a different role in the care environment:

- **The Observational Layer**, responsible for continuous data capture. This includes ambient sensors, mobile tools, wearables, and smart glasses that passively record care interactions, resident behaviors, and environmental context.
- **The Interpretive Layer**, which transforms raw data into actionable insight. AI agents in this layer, like FITL, detect patterns, surface relevant updates, and tailor information to stakeholders in real time.
- **The Strategic Layer**, where institution-wide trends are aggregated and analyzed. Meta-level systems here support planning, resource allocation, and policy decisions by making the needs and preferences of residents visible at scale.

This layered architecture offers one way of approaching the challenge of scaling person-centered care beyond individual interactions. Rather than expecting a single system to serve all users, it distributes the work of interpretation and coordination across levels, mirroring how decisions are already negotiated in long-term care environments. Observations captured at the point of care—through video, sensors, or routine documentation—do not speak for themselves. To become useful, they must first be processed and framed in ways that are accessible and relevant to the people who rely on them. Families, for instance, are not in a position to interpret raw data or navigate unstructured streams of updates. They require information that has already been filtered, contextualized, and made emotionally meaningful. For caregivers too, who are pressed for time, the value of observation lies not in its volume, but in its relevance. Without this initial translation, even rich data can go unused. As we heard in our deployment, caregivers often do not have time to watch a full one-minute video, let alone sift through hours of recorded content or interpret sequences of raw metrics. Without tools to extract and present meaningful moments, this information can be more overwhelming than helpful.

The interpretive layer is designed to support this kind of ongoing sensemaking. It enables asynchronous, low-friction ways of surfacing and sharing care-relevant information, helping stakeholders learn from one another even when they are not directly in conversation. For families, this might mean receiving short, emotionally resonant updates that connect them to their loved one's daily life. For caregivers, it may involve viewing brief contextual notes or summaries that help avoid unnecessary disruptions or repeat mistakes. These interactions—though lightweight—begin to accumulate meaning. Over time, they form a kind of organizational memory that makes patterns and concerns more legible across shifts, roles, and time.

Building on this foundation, the strategic layer supports a broader form of learning. Rather than responding to individual needs or events, this layer looks for recurring signals across residents and across time. Its role is to connect the lived experiences surfaced by the interpretive layer with the planning and decision-making practices of institutional leaders. This includes decisions about staffing models, programming priorities, and long-term investments in care practices. Critically, it is not about removing human judgment but about shaping decision contexts that are more informed by what happens on the ground. In doing so, the strategic layer can help facilities better align their operations with the values of person-centered care—without requiring every individual caregiver or family member to advocate alone.

In proposing this architecture, we draw from principles in multi-agent systems, where higher-order agents coordinate decisions by integrating input from task-specific subagents [34]. While this layered approach is well established in computational domains, its application in dementia care remains limited. Coordination in these settings must account for not only clinical complexity, but also emotional, relational, and infrastructural dynamics that are difficult to model. Recent work has explored the role of AI in diagnostic tasks for dementia [1], but less attention has been given to how such tools might support broader coordination across roles, needs, and time. By outlining this conceptual architecture, we aim to sketch a direction for future systems that move from supporting individual users toward enabling collective action in person-centered care.

5.2 Future Work

Realizing this vision of layered coordination will require sustained inquiry and innovation across multiple fronts. Several stakeholder roles—such as program coordinators and case managers—were not directly supported in our initial deployment, but play critical roles in shaping care continuity and team communication. A future expansion of FITL could better integrate these roles, enabling cross-role coordination and decision support at broader organizational levels.

Long-term care systems will benefit from features that track resident health not only in terms of vital metrics, but also narrative-based trends and emotional well-being. Future systems might support the capture and interpretation of longitudinal data about resident responsiveness, sleep patterns, mood fluctuations, or mobility shifts—helping to identify small changes that matter over time.

We see an opportunity to rethink what constitutes meaningful documentation in dementia care. In our field study, we found that conventional formats (e.g., structured EHR fields, cognitive assessments like the Montreal Cognitive Assessment (MoCA), and checklists of Activities of Daily Living (ADLs)) fall short in capturing the nuanced and often nonlinear experiences of people living with dementia. These measures are ill-equipped to reflect the emotional depth, narrative texture, or momentary lucidity that can be profoundly meaningful to care recipients (See Section 4.3.2), families, and care staff alike.

Without structured ways to account for these human-centered phenomena, the value of person-centered practices risks being discounted in institutional settings. One facility administrator who supported our study, for example, remarked that it was “not worth it” to provide music-based therapy to a particular non-verbal resident. Yet through FITL, we observed subtle but consistent indicators of increased lucidity over repeated sessions—insights that might otherwise have gone unnoticed⁵. While assessing the clinical efficacy of specific therapeutic practices (i.e., those provided by Memcara) was beyond the scope of FITL, the ability to surface and contextualize such moments points to a broader design opportunity. These glimpses of potential therapeutic value show the importance of designing tools that can capture and elevate moments that matter, even when they fall outside conventional documentation or evaluation frameworks.

Future systems might foreground narrative and relational context—what a moment meant, how it was experienced, and how it might guide care going forward. AI can support this shift by detecting and connecting meaningful micro-interactions into richer, evolving storylines that better reflect the lived complexity of dementia.

The ecosystem must also support not only those directly involved in care, but those on the periphery. FITL’s utility could be expanded for family members who are unable to visit regularly, through asynchronous communication, voice-based updates, or adaptive storytelling mechanisms that convey key moments without overwhelming detail. This same infrastructure could support

⁵Unfortunately, staff caregivers told us, this story represents a common occurrence.

augmented reality or wearable technologies to deliver just-in-time information to caregivers on the floor. These directions suggest focusing less on data volume and more on structuring data to support understanding, continuity, and dignity. The path forward will require tools that operate across tiers of care: from moment-to-moment support to long-term system transformation.

5.3 Navigating Tradeoffs in Scaling Care Systems

While our deployment surfaced a compelling vision, it also exposed a set of persistent tensions that must be more fully reckoned with as systems like FITL scale. Even in a limited deployment, we encountered tradeoffs between automation and manual oversight, and between transparency and privacy. Staff appreciated the reduced burden of documentation but expressed concern about how their work might be interpreted. Families valued increased visibility into care, but at times felt overwhelmed or unsure how to engage with what they saw. These frictions reveal deeper questions of trust, discretion, and control—questions that become only more complex as care systems grow more data-driven.

These are not problems with definitive solutions, but ongoing negotiations. Designing for scale means designing for stewardship—supporting continuous calibration between institutional needs and the lived realities of caregiving. In this light, the tensions we observed are not side effects of innovation but core design considerations for any future ecosystem. FITL demonstrates that AI can be meaningfully embedded in the everyday practices and relationships that shape care. However, realizing its potential at scale will require more than technical refinement. It calls for sustained attention to the human, organizational, and ethical structures in which these tools operate. The opportunity ahead is not just to expand what care systems can observe, but to strengthen what they are able to understand and sustain.

6 Conclusion

Through an iterative design and development process, we created and evaluated Family In The Loop, an AI-driven web application to support person-centered dementia caregiving. Building on prior studies centered on caregiving staff, this work explores a new, often-overlooked direction: empowering care-recipients' families through a dedicated, family-focused portal that is deeply integrated into existing caregiver workflows and existing EHR systems. Our work seeks to improve the understanding of how technology can support information sharing and synthesis between families and caregivers. We follow a human-centered, collaborative design approach that is focused on deeply understanding the needs of facility staff, PLWD, and family members. We evaluate our final prototype through a deployment study and present rich qualitative findings from in-person observations and interviews. Our findings suggest promising improvements in family involvement and trust, and in cross-role communication, which in turn seem to bolster the joint capacity of families and staff in long-term care—as a complex adaptive system—to provide more person-centered care. Finally, we reflect on the limitations of current systems and propose a layered view of data-driven, person-centered care as a path forward. To foster care ecosystems that are both data-rich and deeply human-centered, we offer this framework as a roadmap for AI-enabled systems that connect frontline caregiving with institutional decision-making.

7 Acknowledgments

We extend our deepest gratitude to the care recipients, their family members, professional caregivers, facility administrators, Memcara staff—in particular Markus Nottelmann, and other project collaborators who made this research possible. We also thank the undergraduate, master's, and PhD students who contributed to the project – especially Bansharee Ireen, Sophia Moore, Pape Sow Traore, Vafa Batool, and Sid Patil. We also thank Melody Yin and Sab Garduño for reviewing this paper. Lastly, we acknowledge the support of the Dartmouth PhD Innovation Program.

References

- [1] Amirhossein Eslami Andargoli, Nalika Ulapan, Tuan Anh Nguyen, Nadeem Shukat, John Zelcer, and Nilmini Wickramasinghe. 2024. Intelligent decision support systems for dementia care: A scoping review. *Artificial Intelligence in Medicine* (2024), 102815.
- [2] George L Anesi and Meeta Prasad Kerlin. 2021. The impact of resource limitations on care delivery and outcomes: routine variation, the coronavirus disease 2019 pandemic, and persistent shortage. *Current opinion in critical care* 27, 5 (2021), 513–519.
- [3] Ramona Backhaus, Linda JM Hoek, Erica de Vries, Jolanda CM van Haastregt, Jan PH Hamers, and Hilde Verbeek. 2020. Interventions to foster family inclusion in nursing homes for people with dementia: a systematic review. *BMC geriatrics* 20, 1 (2020), 434.
- [4] Rachel Barken and Ruth Lowndes. 2018. Supporting family involvement in long-term residential care: Promising practices for relational care. *Qualitative health research* 28, 1 (2018), 60–72.
- [5] Michael Bauer and Rhonda Nay. 2011. Improving family–staff relationships in assisted living facilities: the views of family. *Journal of Advanced Nursing* 67, 6 (2011), 1232–1241. <https://doi.org/10.1111/j.1365-2648.2010.05575.x>
- [6] Claus Bossen and Randi Markussen. 2010. Infrastructuring and ordering devices in health care: Medication plans and practices on a hospital ward. *Computer Supported Cooperative Work (CSCW)* 19 (2010), 615–637.
- [7] Lindsey Brett, Victoria Traynor, and Paul Stapley. 2016. Effects of physical exercise on health and well-being of individuals living with a dementia in nursing homes: a systematic review. *Journal of the American Medical Directors Association* 17, 2 (2016), 104–116.
- [8] Sonya Brownie and Susan Nancarrow. 2013. Effects of person-centered care on residents and staff in aged-care facilities: a systematic review. *Clinical interventions in Aging* (2013), 1–10.
- [9] Karabi C. Bezboruah, Darla Paulson, and Jason Smith. 2014. Management attitudes and technology adoption in long-term care facilities. *Journal of health organization and management* 28, 3 (2014), 344–365.
- [10] Marie-Andrée Cadieux, Linda J Garcia, and Jonathan Patrick. 2013. Needs of people with dementia in long-term care: a systematic review. *American Journal of Alzheimer's Disease & Other Dementias* 28, 8 (2013), 723–733.
- [11] Áine Carroll, Claire Collins, Jane McKenzie, Diarmuid Stokes, and Andrew Darley. 2023. Application of complexity theory in health and social care research: a scoping review. *BMJ open* 13, 3 (2023), e069180.
- [12] Mary W Chaffee and Margaret M McNeill. 2007. A model of nursing as a complex adaptive system. *Nursing outlook* 55, 5 (2007), 232–241.
- [13] You Chen, Christoph U Lehmann, and Bradley Malin. 2024. Digital information ecosystems in modern care coordination and patient care pathways and the challenges and opportunities for AI solutions. *Journal of Medical Internet Research* 26 (2024), e60258.
- [14] Josh Cherian, Samantha Ray, Thomas Mernar, Paul Taele, Helen Mach, Jung In Koh, Ping Ye, and Tracy Hammond. 2024. A Step Toward Better Care: Understanding What Caregivers and Residents in Assisted Living Facilities Value in Health Monitoring Systems. *Proceedings of the ACM on Human-Computer Interaction* 8, CSCW1 (2024), 1–29.
- [15] Design Council. 2021. *What is the framework for innovation? Design Council's evolved Double Diamond*. Design Council.
- [16] Nils Dahlbäck, Arne Jönsson, and Lars Ahrenberg. 1993. Wizard of Oz studies: why and how. In *Proceedings of the 1st international conference on Intelligent user interfaces*. 193–200.
- [17] Ayelet Dassa. 2018. “Opening Our Time Capsule”—creating an individualized music and other memory cues database to promote communication between spouses and people with dementia during visits to a nursing home. *Frontiers in Medicine* 5 (2018), 215.
- [18] Suzanne M Dyer, Stephanie L Harrison, Kate Laver, Craig Whitehead, and Maria Crotty. 2018. An overview of systematic reviews of pharmacological and non-pharmacological interventions for the treatment of behavioral and psychological symptoms of dementia. *International psychogeriatrics* 30, 3 (2018), 295–309.
- [19] K Anders Ericsson and Herbert A Simon. 1980. Verbal reports as data. *Psychological review* 87, 3 (1980), 215.
- [20] Sam Fazio, Douglas Pace, Janice Flinner, and Beth Kallmyer. 2018. The fundamentals of person-centered care for individuals with dementia. *The gerontologist* 58, suppl_1 (2018), S10–S19.
- [21] Hannah Flamm. 2018. “They Want Docile”: How Nursing Homes in the United States Overmedicate People with Dementia. <https://www.hrw.org/report/2018/02/05/they-want-docile/how-nursing-homes-united-states-overmedicate-people-dementia>
- [22] Jane Fossey, Clive Ballard, Edmund Juszcak, Ian James, Nicola Alder, Robin Jacoby, and Robert Howard. 2006. Effect of enhanced psychosocial care on antipsychotic use in nursing home residents with severe dementia: cluster randomised trial. *Bmj* 332, 7544 (2006), 756–761.
- [23] Ashvin Gandhi, Huizi Yu, and David C Grabowski. 2021. High nursing staff turnover in nursing homes offers important quality information: study examines high turnover of nursing staff at US nursing homes. *Health Affairs* 40, 3 (2021),

- 384–391.
- [24] Joseph E Gaugler and Lauren L Mitchell. 2022. Reimagining family involvement in residential long-term care. *Journal of the American Medical Directors Association* 23, 2 (2022), 235–240.
 - [25] Patricia R Gilman. 2021. Complex adaptive systems: a framework for an integrated chronic care model. *Advances in Nursing Science* 44, 4 (2021), 330–339.
 - [26] Emily Haesler, Michael Bauer, and Rhonda Nay. 2007. Staff–family relationships in the care of older people: a report on a systematic review. *Research in nursing & health* 30, 4 (2007), 385–398.
 - [27] Janine K Hayward, Charlotte Gould, Emma Palluotto, Emily C Kitson, and Aimee Spector. 2023. Family involvement with care homes following placement of a relative living with dementia: a review. *Ageing & Society* 43, 7 (2023), 1530–1575.
 - [28] Julia Hermann, Aiden Danny Mäder, Ann-Kathrin Kubullek, Celine-Chiara Hey, and Aysegül Dogangün. 2022. (Intelligent) Technical Systems in Elderly Care: The Caregivers Perspective. In *Proceedings of the 34th Australian Conference on Human-Computer Interaction*. 81–87.
 - [29] Nicole TM Hill, Loren Mowszowski, Sharon L Naismith, Verity L Chadwick, Michael Valenzuela, and Amit Lampit. 2017. Computerized cognitive training in older adults with mild cognitive impairment or dementia: a systematic review and meta-analysis. *American Journal of Psychiatry* 174, 4 (2017), 329–340.
 - [30] Anna Sigridur Islind, Helena Vallo Hult, Victoria Johansson, Eva Angenete, and Martin Gellerstedt. 2021. Invisible work meets visible work: infrastructuring from the perspective of patients and healthcare professionals. In *54th Hawaii International Conference on System Sciences (HICSS9, Tuesday, January 5, 2021 to Friday, January 8, 2021)*. Hawaii International Conference on System Sciences, 3556–3565.
 - [31] Ann Kolanowski, Marie Boltz, Elizabeth Galik, Laura N Gitlin, Helen C Kales, Barbara Resnick, Kimberly S Van Haitsma, Amy Knehans, Jane E Sutterlin, Justine S Sefcik, et al. 2017. Determinants of behavioral and psychological symptoms of dementia: A scoping review of the evidence. *Nursing outlook* 65, 5 (2017), 515–529.
 - [32] Torsten Kratz. 2017. The diagnosis and treatment of behavioral disorders in dementia. *Deutsches Ärzteblatt International* 114, 26 (2017), 447.
 - [33] Karan S Kverno, Betty S Black, Marie T Nolan, and Peter V Rabins. 2009. Research on treating neuropsychiatric symptoms of advanced dementia with non-pharmacological strategies, 1998–2008: a systematic literature review. *International psychogeriatrics* 21, 5 (2009), 825–843.
 - [34] Naomi Ehrich Leonard, Anastasia Bizyaeva, and Alessio Franci. 2024. Fast and flexible multiagent decision-making. *Annual Review of Control, Robotics, and Autonomous Systems* 7 (2024).
 - [35] David A Lindeman, Katherine K Kim, Caroline Gladstone, and Ester Carolina Apesoa-Varano. 2020. Technology and caregiving: emerging interventions and directions for research. *The Gerontologist* 60, Supplement_1 (2020), S41–S49.
 - [36] Jill Manthorpe and Kritika Samsi. 2016. Person-centered dementia care: current perspectives. *Clinical interventions in ageing* (2016), 1733–1740.
 - [37] John Meiklejohn, Catherine Phillips, M Lee Freedman, Mary Lee Griffin, Gina Biegel, Andy Roach, Jenny Frank, Christine Burke, Laura Pinger, Geoff Soloway, et al. 2012. Integrating mindfulness training into K-12 education: Fostering the resilience of teachers and students. *Mindfulness* 3 (2012), 291–307.
 - [38] Dylan Edward Moore, Songyun Tao, Sophia RR Moore, Christina Sapp Tadin, Dio Tadin, Brian Morgan, and Elizabeth L Murnane. 2025. *CareInsights: AI-Enabled Infrastructure for Person-Centered Dementia Care in Resource-Constrained Facilities*. Thayer School of Engineering, Dartmouth College. <https://empowerlab.dartmouth.edu/projects/care-insights>
 - [39] Tamara Mujirishvili, Caterina Maidhof, Francisco Florez-Revuelta, Martina Ziefle, Miguel Richart-Martinez, and Julio Cabrero-García. 2023. Acceptance and privacy perceptions toward video-based active and assisted living technologies: scoping review. *Journal of Medical Internet Research* 25 (2023), e45297.
 - [40] Rob Munro. 2021. *Human-in-the-Loop Machine Learning: Active learning and annotation for human-centered AI*. Manning.
 - [41] Jakob Nielsen. 1994. *Usability engineering*. Morgan Kaufmann.
 - [42] World Health Organization. 2023. Dementia. <https://www.who.int/news-room/fact-sheets/detail/dementia> Last updated September 20, 2023.
 - [43] World Health Organization et al. 2015. *WHO global strategy on people-centred and integrated health services: interim report*. Technical Report. World Health Organization.
 - [44] World Health Organization et al. 2021. Global status report on the public health response to dementia. (2021).
 - [45] Carol Ann Podgorski, Sharon D Anderson, and Jasneet Parmar. 2021. A biopsychosocial-ecological framework for family-farmed dementia care. *Frontiers in Psychiatry* 12 (2021), 744806.
 - [46] Gloria Puurveen, Jennifer Baumbusch, and Preet Gandhi. 2018. From family involvement to family inclusion in nursing home settings: A critical interpretive synthesis. *Journal of family nursing* 24, 1 (2018), 60–85.
 - [47] Peter Pype, Fien Mertens, Fleur Helewaut, and Demi Krystallidou. 2018. Healthcare teams as complex adaptive systems: understanding team behaviour through team members’ perception of interpersonal interaction. *BMC health services*

- research 18 (2018), 1–13.
- [48] Cheryl Rathert, Julie Brandt, and Eric S Williams. 2012. Putting the ‘patient’ in patient safety: a qualitative study of consumer experiences. *Health Expectations* 15, 3 (2012), 327–336.
 - [49] R Colin Reid, Neena L Chappell, and Jessica A Gish. 2007. Measuring family perceived involvement in individualized long-term care. *Dementia* 6, 1 (2007), 89–104.
 - [50] Regina A Shih, Thomas W Concannon, Jodi L Liu, and Esther M Friedman. 2014. Improving dementia long-term care: A policy blueprint. *Rand health quarterly* 4, 2 (2014), 2.
 - [51] Mark Snowden, Kersten Sato, and Peter Roy-Byrne. 2003. Assessment and treatment of nursing home residents with depression or behavioral symptoms associated with dementia: a review of the literature. *Journal of the American Geriatrics Society* 51, 9 (2003), 1305–1317.
 - [52] Carolyn Snyder. 2003. *Paper prototyping: The fast and easy way to design and refine user interfaces*. Morgan Kaufmann.
 - [53] George Sun and Yi-Hui Zhou. 2023. AI in healthcare: navigating opportunities and challenges in digital communication. *Frontiers in Digital Health* 5 (2023), 1291132.
 - [54] Lorainne Tudor Car, Dhakshenya Ardhithy Dhinakaran, Bhone Myint Kyaw, Tobias Kowatsch, Shafiq Joty, Yin-Leng Theng, and Rifat Atun. 2020. Conversational agents in health care: scoping review and conceptual analysis. *Journal of medical Internet research* 22, 8 (2020), e17158.
 - [55] Queen Utley-Smith, Cathleen S Colón-Emeric, Deborah Lekan-Rutledge, Natalie Ammarell, Donald Bailey, Kirsten Corazzini, Mary L Piven, and Ruth A Anderson. 2009. Staff perceptions of staff-family interactions in nursing homes. *Journal of Aging Studies* 23, 3 (2009), 168–177.
 - [56] Myrra Vernooij-Dassen, Emmelyne Vasse, Sytse Zuidema, Jiska Cohen-Mansfield, and Wendy Moyle. 2010. Psychosocial interventions for dementia patients in long-term care. *International Psychogeriatrics* 22, 7 (2010), 1121–1128.
 - [57] Lu Wang, Diva Smriti, Hao Yuan, and Jina Huh-Yoo. 2024. Artificial Intelligence Systems for Supporting Informal Caregivers of People Living with Alzheimer’s Disease or Related Dementias: A Systematic Review. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. 1–11.
 - [58] Jaime Williams, Thomas Hadjistavropoulos, Omeed O Ghandehari, XUE Yao, and Lisa Lix. 2015. An evaluation of a person-centred care programme for long-term care facilities. *Ageing & Society* 35, 3 (2015), 457–488.
 - [59] Kristine N Williams, Yelena Perkhounkova, Ying-Ling Jao, Ann Bossen, Maria Hein, Sophia Chung, Anne Starykowicz, and Margaret Turk. 2018. Person-centered communication for nursing home residents with dementia: Four communication analysis methods. *Western journal of nursing research* 40, 7 (2018), 1012–1031.
 - [60] Shumenghui Zhai, Frances Chu, Minghui Tan, Nai-Ching Chi, Teresa Ward, and Weichao Yuwen. 2023. Digital health interventions to support family caregivers: An updated systematic review. *Digital health* 9 (2023), 20552076231171967.

8 Appendix

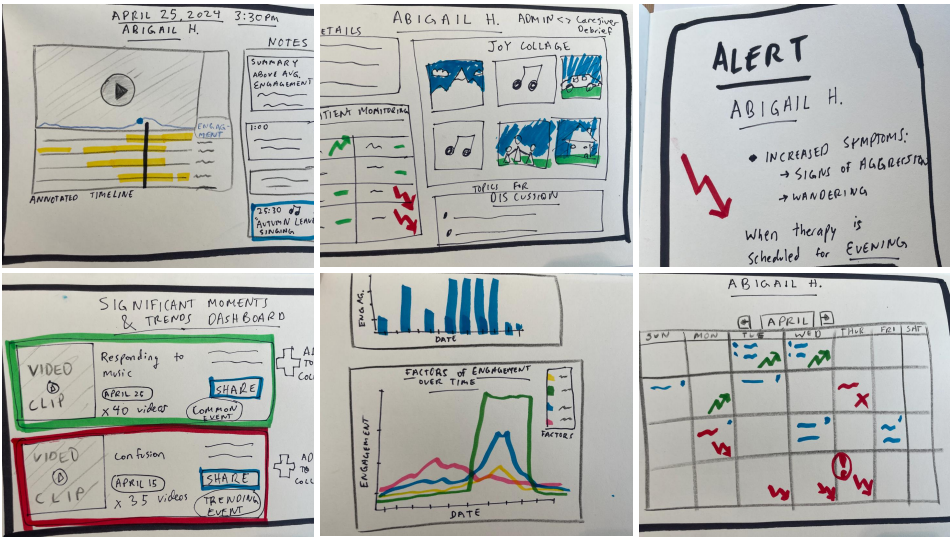


Fig. 5. Excerpts from our early low-fidelity prototypes. All user content shown here is mocked data.

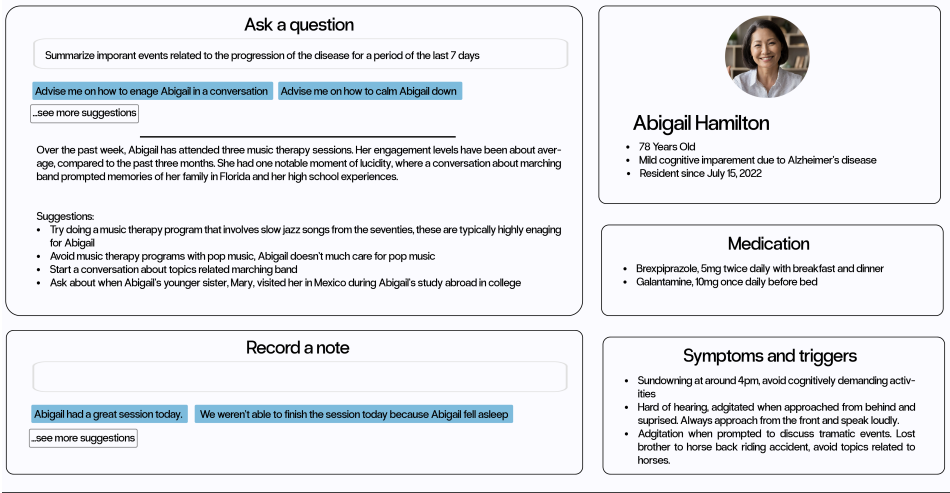


Fig. 6. Excerpt from our medium-fidelity prototypes. All user content shown here is mocked data.

8.1 Detailed system overview

8.1.1 Onboarding. FITL includes an onboarding page featuring walkthrough videos created by the authors to guide users through each view. First-time users are automatically directed to this onboarding page upon opening the app. No account creation or manual data entry is required; user identity, role, and care recipient information are seamlessly integrated through single sign-on via Memcara's existing platform.

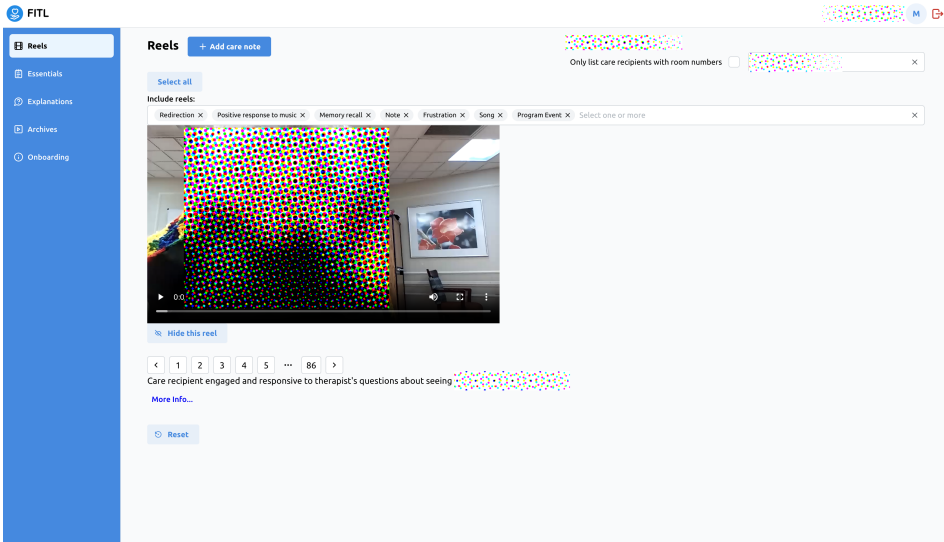


Fig. 7. A screenshot of Reels View. All personally identifiable information has been pixelated.

8.1.2 Reels View. Reels View serves as the primary entry point for families, reflecting their preference for regular, emotionally meaningful updates. Updates come in the form of short video clips, or "reels", drawn from structured therapeutic interactions between staff and residents, provide a low-effort way to remain connected to the care experience, particularly when in-person observation is not feasible.

Video capture for these 1:1 sessions occurs passively by leveraging the Memcara platform's existing tooling. Caregivers use a fixed-position iPad⁶ already in place for delivering assistive recorded content (e.g., music videos with generic memory related question prompts) during therapy—which continuously faces the care recipient, allowing sessions to be captured with minimal effort. Caregivers can initiate video capture with a single tap, and the device reliably records a well-framed interaction without requiring additional setup or repositioning. This design minimizes the burden on care staff, who can document meaningful moments in ways that align with their existing workflows and without disrupting the flow of the session.

While many reels highlight positive moments—such as a resident singing along to music, recognizing a photograph, or laughing during a conversation—Reels View may also surface more difficult or complex episodes, such as a moment of confusion redirected effectively by a caregiver or a resident expressing frustration during a memory activity. These moments are included when

⁶Currently, our partner facilities have only a few iPads that are shared by the staff, these are repositioned each time a 1-1 staff-care recipient interaction.

they are marked by atypical behaviors or salient caregiver responses, offering families greater transparency into the realities of dementia care while supporting reflection and learning. Reels are dynamically selected by the CareInsights backend.

Each reel is paired with an AI-generated summary synthesized by the CareInsights backend, such as “Resident identifies their daughter in a photo and talks about a family trip,” “Initially resists participating in memory exercise but begins to engage after caregiver prompts with familiar music,” or “Caregiver calmly redirects agitation during afternoon confusion episode.” These captions aim to foreground the emotional or narrative significance of the moment without requiring families to interpret raw footage. The “More Info” button beneath each summary transitions the user to Explanations View (see Section 8.1.3), where the system pre-populates a clarifying question (e.g., “What prompted the change in mood during the therapy session on <Date:Time>?”) and returns an explanation grounded in the surrounding care record.

To help families manage emotional load, Reels can be filtered using a horizontal bar that groups clips by topic or emotional tone—for instance, “Positive response to music,” “Redirection,” or “Memory recall.” This allows families to avoid—or stay updated on—content that may be emotionally difficult. Individual clips can also be hidden, and doing so updates the backend’s clip selection algorithm, deprioritizing similar moments in future Reels by updating the backend curation logic.

Reels View is embedded within a shared data infrastructure designed to accommodate iterative feedback. During prototyping, we found that families’ responses to clips varied widely—not only in emotional interpretation, but in what was considered helpful or appropriate to share. Reels includes an “Add Care Note” button, allowing users to comment on moments with reflections or clarifications (e.g., “*He always did love Kathleen. He used to hum her favorite song, Vienna, when he missed her.*”). These notes are appended to the CareInsights data layer and trigger re-synthesis of the AI-generated summary, influencing both the displayed caption, the system’s evolving model of which moments to surface in the future, and AI-generated data on other views.

Reels is intentionally lightweight. The interface presents a paginated feed of video clips with thumbnails emphasizing facial expressions or gestures. All content is drawn from structured therapy contexts—not continuous surveillance—and administrative controls in Archives View allow staff to review or restrict what recordings can be shared at a per-session level⁷.

8.1.3  Explanations View. The Explanations View enables both families and staff to request clarification, context, or elaboration on care-related observations. These questions may pertain to recent interactions (e.g., “Describe the most recent care activities”), behavioral patterns (“Has he been showing signs of agitation now more than usual?”), or care practices (“What usually helps bring out moments of lucidity?”).

When a user enters the view—either via a drill-down link or directly from the navigation sidebar—they are presented with a suggested question and answer based on the preceding interaction, meaning the view is never without content⁸. The existing prompt can then be updated.

To promote transparency without overwhelming users, FITL’s response to the query appears as a short paragraph with the user’s question at the top. Reference icons embedded in the text link directly to supporting documentation, such as text-based notes (i.e., from EHR systems or staff/family feedback) () or timestamped video clips (). For example: “*Caregiver observed increased agitation, music therapy involving Andrea Bocelli was used as a calming technique.*” () Clicking on

⁷Placing ALL privacy controls in the Archives View was, conveniently, both more manageable for staff in practice and allows for more flexibility when making dynamic content updates.

⁸In our early prototyping, we found that users got confused by/unsure how to use views that felt “empty,” having a default question and answer showing at all times gave users some context for both how to use the view as well as mentally triggered them to think of additional questions to ask.

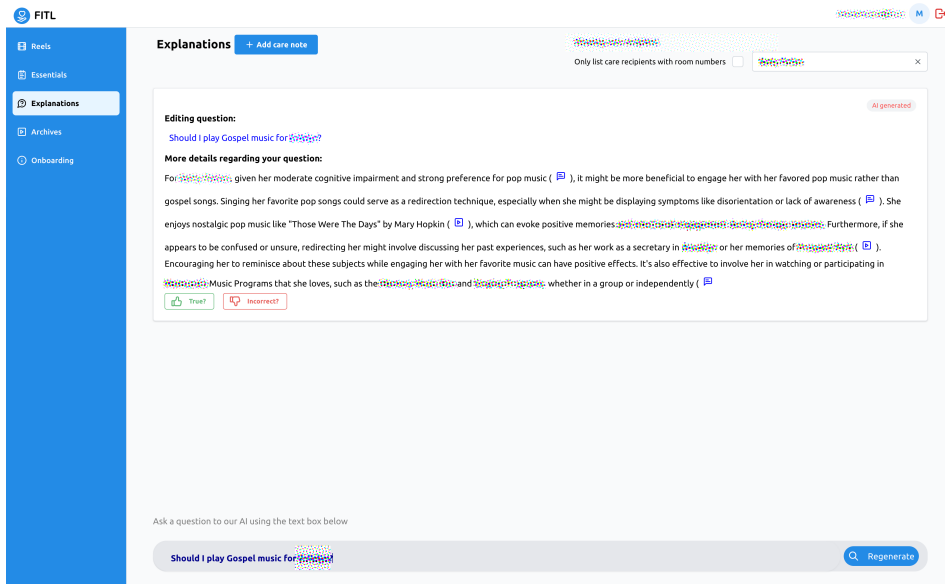


Fig. 8. Explanations View. All personally identifiable information has been pixelated.

these citations help users do a deep dive into the underlying data, to verify how a conclusion was reached.

Like the real-world practices it emulates—such as phone calls or in person exchanges—Explanations View supports a conversational interaction model. Users can submit follow-up or rephrased questions, prompting the system to regenerate responses using updated query parameters. This enables asynchronous, lightweight dialogue that mirrors the kinds of clarifications families might seek during visits, or that staff routinely exchange as part of everyday coordination.

Like Reels and Essentials, Explanations View includes an "Add Care Note" button for free response. However, at users' request for a faster way to validate or reject responses for the system to learn we also included corresponding "Useful" and "Not Useful" buttons. On the backend, feedback is tied to the underlying records and prompts the system to reprocess its explanation (and other views that reference the updated info) accordingly. For example, a family member might write, "That's not a good redirection strategy for Lucy—she responds better to music and mentions of her sister, Rachel." This input is saved to the CareInsights data layer and influences future outputs involving the resident. A similar (though, less precise⁹ result) could be achieved by clicking on the "Incorrect?" button. Over time, these mechanisms help refine the system's understanding of individual preferences and embed practical care strategies into the shared knowledge base.

⁹While the initial response would be updated to avoid making the noted claim, without the added context of a free response input, the new result could take a more generic sound. For this reason, users took the time to use the "Add Care Note" button, encouraged by the sense that FITL was learning.

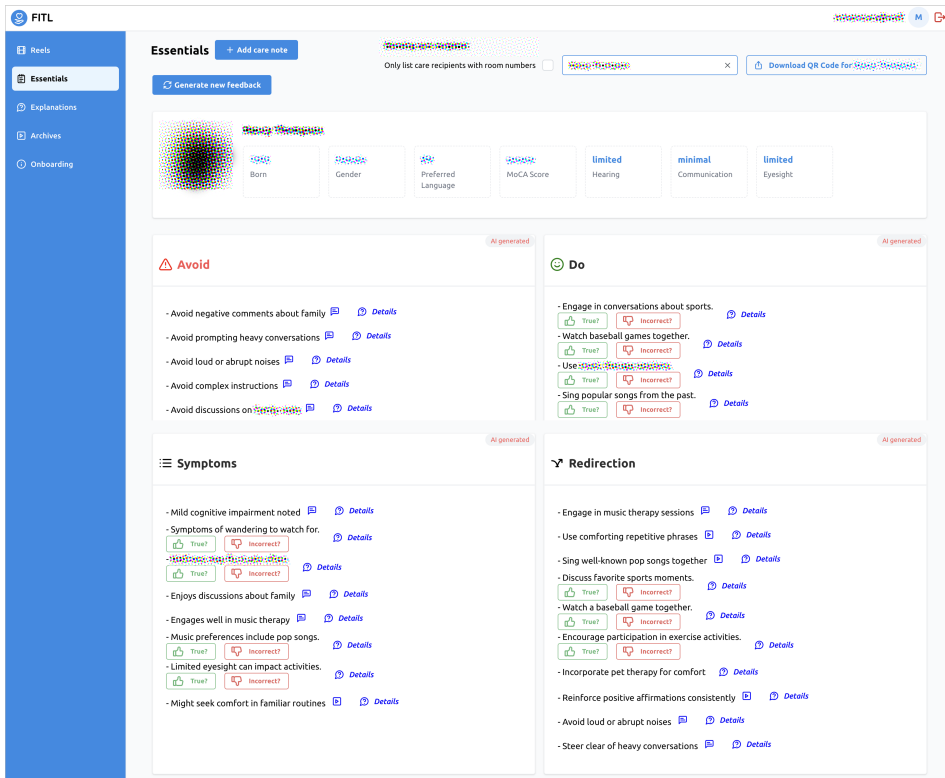


Fig. 9. A screenshot of Essentials View. All personally identifiable information has been pixelated.

8.1.4 Essentials View. We created the Essentials View after seeing how staff often carried informal “cheat sheets” summarizing a resident’s most important traits—such as how they tend to behave on a typical day, what strategies can defuse their agitation, and which activities spark engagement or cause distress. By transforming those hand-scrawled bullet points into a digitally maintained list, we help ensure that every member of the care team, as well as family members with the right permissions, can see the same up-to-date guidance. Following staff members’ suggestions, Essentials View is split into five boxes:

- (1) A top-level, “Profile” box with the most important info (i.e., name, preferred language, communication capabilities, hearing capabilities, etc.),
- (2) “⚠️ Avoid” (i.e., activities or topics to avoid),
- (3) “😊 Do” (i.e., activities or topics to engage in),
- (4) “☰ Symptoms” (i.e., common dementia-related behaviors and comorbidities exhibited by this individual), and
- (5) “↪️ Redirection” (i.e., effective strategies for redirecting the individual).

The four bottom boxes house a brief set of bullet points summarizing insights that are identified as relevant to delivering person-centered care.

From a technical perspective, this is handled by querying the backend with a corresponding question—“What behaviors are typical?”, “How do we de-escalate them?”, “What helps them engage?”, “What should we avoid?”—and specifying that the format of the response should be a

bulleted list. The backend pipeline gathers data relevant (i.e., processed text summaries of videos, EHR system data, and user feedback) to each question, synthesizes a response, and then formats that response into the requested bulleted format. The template passed into the API specifically limits the word count per bullet, so that staff, or family, can understand the most important takeaways at a glance. This is the same API used for generating responses on Explanations View, however, in that case a different template is used to achieve a longer, narrative form response. As with Essentials View, claims here are supported by citations that function as transitions to Archives View.

A "Details" button accompanying each bullet was included at the request of staff who wanted as a way to quickly navigate to the Explanations View and pre-populate a related follow-up question to the bullet (i.e., "tell me more about <bullet content>"). This was important because bullets are necessarily limited in their ability to convey nuance, and most users expressed a need for an (ideally, "one-click") option to get more "details" on demand.

To address moments of system uncertainty, we included a minimal feedback mechanism: thumbs-up and thumbs-down icons appear next to certain bullet points in the Essentials View. These icons are not shown universally. Rather, they appear only when the system detects that a bullet lacks explicit citations—usually in cases where few structured records exist to back the claim¹⁰. This approach emerged from user feedback: staff and families wanted a way to quickly respond to suggestions that felt too generic or possibly misleading, but they also recognized that some of these uncited suggestions were helpful and should not be removed outright.

The presence of thumbs icons signals that the system itself considers the statement provisional. Users can click thumbs-up or thumbs-down and optionally add a brief comment. If approved, a system-generated note is stored in the data layer (e.g., "Caregiver [Name] confirmed this strategy on [Date]"), which serves as a supporting record. The item is then considered "certain" and the icons are removed in future views. This balances the risk of misleading content with the potential value of AI-generated suggestions, especially for care recipients with limited data on record.

In contrast, when a bullet is already tied to an existing care note or transcript reference, errors require more than a vote of confidence to resolve. These cases demand a higher degree of accountability: the underlying record must be corrected or augmented. To support this, we direct users to the "Add Care Note" button for submitting corrections. Requiring written input in these cases ensures that edits are well-documented, supporting downstream data quality across all views.

When feedback is submitted, the system updates immediately, allowing users to see how their input was interpreted. If the update fails to address the issue, the visible output provides a concrete prompt for refinement—another care note can be added to clarify or correct what the system inferred. We observed this iterative model as critical to maintaining data accuracy, while also making system behavior transparent. In more involved cases, users can navigate to the source record and directly edit it—ensuring that the system remains grounded in a reviewable, collectively maintained data layer.

¹⁰In practice, we found that bullets without citations only appear for care recipients with very few recorded videos (e.g., fewer than a dozen, though this can vary). In these cases, the generated content—but particularly uncertain bullet points—tends to be generic and may be less helpful or even misleading. We observed that once a care recipient has more than ten videos on record, the content in this view becomes more grounded in observed events and behavioral patterns. At that point, offering quick feedback becomes less appropriate, as edits typically require more nuanced reasoning and justification.

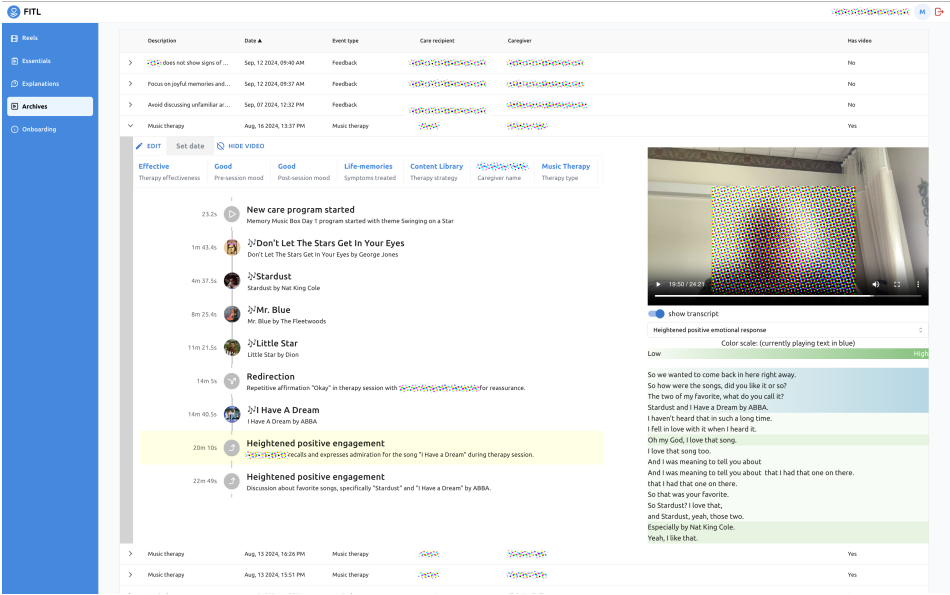


Fig. 10. A screenshot of Archives View. The header and the “Add Care Note” button are not shown (to capture more of the other features in-frame), but appear at the top of the page. All personally identifiable information has been pixelated.

8.1.5  *Archives View.* Archives View serves as the canonical, full-fidelity interface for interacting with a resident’s longitudinal care record. Inspired by the paper-based tracking systems observed among “super-user” facility staff, the view was designed to replicate the format of hand-curated binders that included timelines, annotations, and program notes. This design—vertical, expandable, and chronologically ordered—offers an at-a-glance overview of recent care and the option to drill down into fine-grained documentation. It functions in FITL as a centralized reference interface, supporting both record verification and the editing of underlying care records that drive updates across the rest of the views (which show dynamically generated content).

To support these needs, Archives View displays both structured and multimedia documentation. Text-only records (e.g., shift logs, staff notes, family-submitted observations) appear in the expandable list alongside video records. When a record is derived from a video recording, a set of embedded subviews becomes available (See Figure 10): the video is shown alongside a transcript, with dynamic highlighting based on AI-generated annotations. These annotations identify caregiver-relevant behaviors like memory recall, emotional engagement, or behavioral deviations. Highlighting can be toggled to emphasize different engagement types, and clicking on transcript lines or timeline markers jumps the video to the associated moment.

Adjacent to the transcript, a vertically stacked timeline visualizes system-detected moments of salience—such as caregiver responses, family-relevant clips (used in Reels View), and successful redirection. When music is detected, album art appears inline. Like the transcript, the timeline serves as an interactive index for navigating the video. It highlights the current timestamp and allows users to jump to specific moments with a single click. On the backend, the timeline is generated using timestamped logs from Memcara’s platform, which provides a continuous stream of events corresponding to stimuli presented during therapy sessions—such as images, music, or verbal prompts. Because the platform records video concurrently with this structured event stream,

the system can align what the care recipient is seeing or hearing at any given moment with how they behave or respond in the footage. This integration enables the system to reason about potential causal relationships—for example, identifying whether an observed reaction (such as memory recall or emotional shift) coincided with a specific song or visual prompt. These aligned timelines allow caregivers and families to more easily understand not just what happened, but why—supporting clinical interpretation, family insight, and more targeted care planning. Taken together, these elements offer a comprehensive picture of each recorded session, dramatically reducing the time required to understand what happened—and solving a common complaint raised by staff during early prototyping: no one had time to watch entire videos.

Editing tools in Archives View are designed to support both granular and systemic updates. Staff can revise the AI-generated transcript (e.g., correcting a misinterpreted word), edit or re-label a record's metadata, or add entirely new entries. These updates propagate across Reels, Essentials, and Explanations Views, ensuring that all representations of a resident's care remain consistent. Because all generated content in FITL maintains references to its source records—either explicitly or internally—the system uses targeted re-rendering to update only content affected by a change.

This propagation logic—driven by dependency-tracking and view-aware update rules—ensures that even lightweight feedback (like a thumbs-down click) can cascade across views¹¹. In cases where ambiguity arises (i.e., "Add a Care Note" on Essentials View), page context (i.e., what citations are present) helps narrow down which record is most likely being referenced, enabling even brief notes to be actionable. Manual edits remain visible as part of the record's revision history, making feedback legible and accountable.

Visibility controls were designed to be intentionally simple. Staff can set default permissions—e.g., defaulting new content to "Staff-only" or "Family-visible"—or adjust them record by record. During late-stage prototyping, two partner sites informed us that EHR-linked media and transcripts were sometimes subject to "on-prem only" access policies. In response, we added geofencing: when enabled, the system checks the user's GPS location before loading restricted records. This feature provided staff with additional reassurance that sensitive content would remain within facility bounds when required, balancing accountability and protection.

Archives View serves as the foundation of the FITL system, providing a persistent and editable record of care from which all other views derive their content. While other views offer synthesized and filtered perspectives, Archives gives users full access to the raw data and tools to shape what the system learns. By combining editable records, media navigation, and targeted propagation of updates, Archives promotes a high-trust documentation experience that reflects the constraints, values, and practices of long-term dementia care.

Received October 2024; revised April 2025; accepted August 2025

¹¹Feedback submitted through actions like "Thumbs-down" or "Add a Care Note" is recorded as a new entry in Archives View and linked to the specific record it references—forming a traceable graph of relationships. This structure enables FITL to backpropagate updates that originate as feedback on other views (rather than direct edits on Archives View), allowing it to revise dependent content while maintaining a transparent and traceable history of changes.